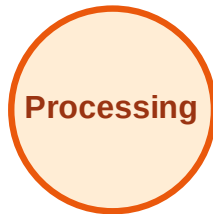


BFS Traversal

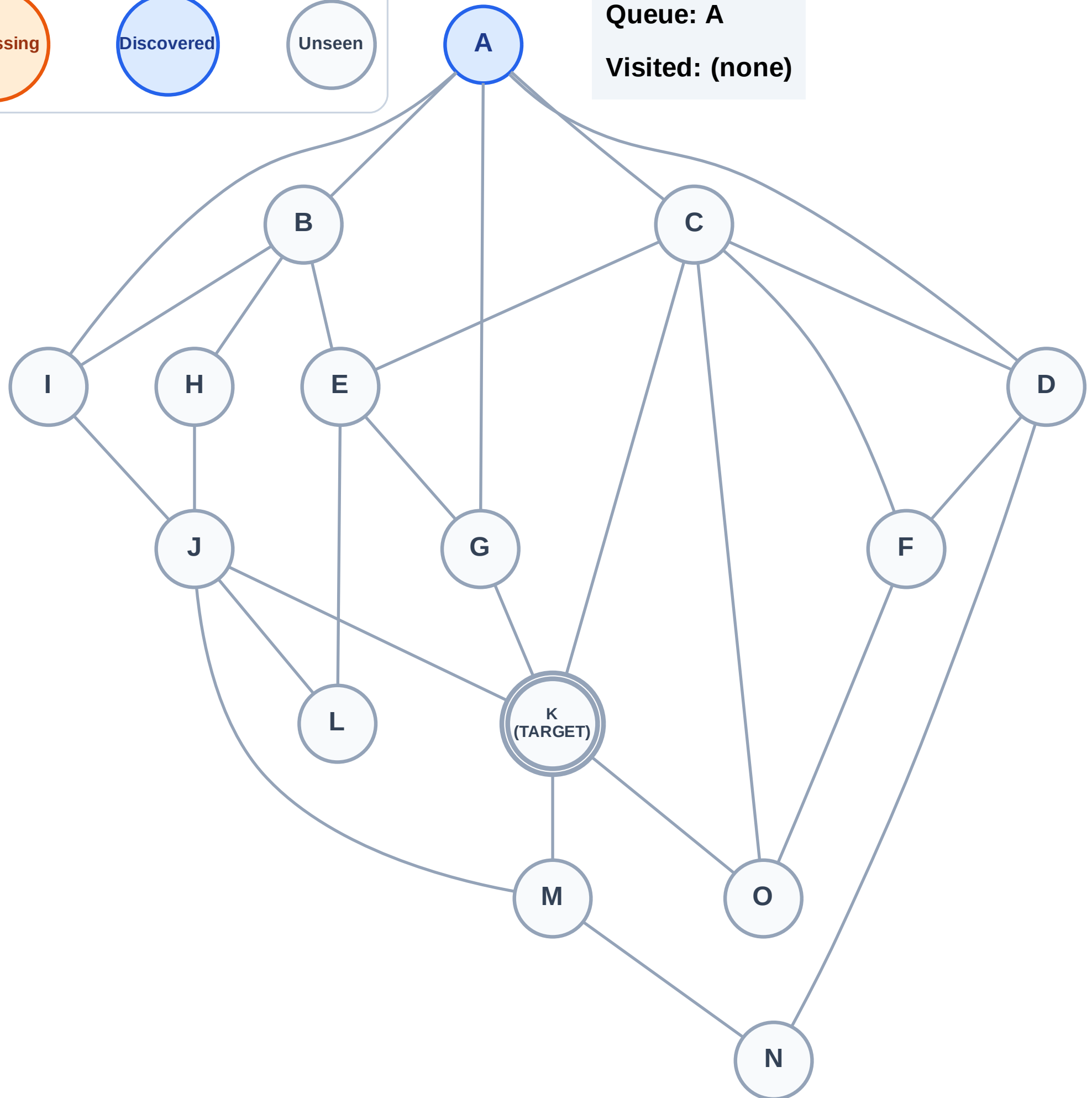
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

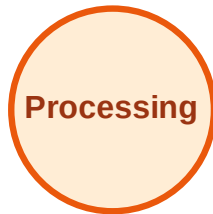
Visited: (none)



BFS Traversal

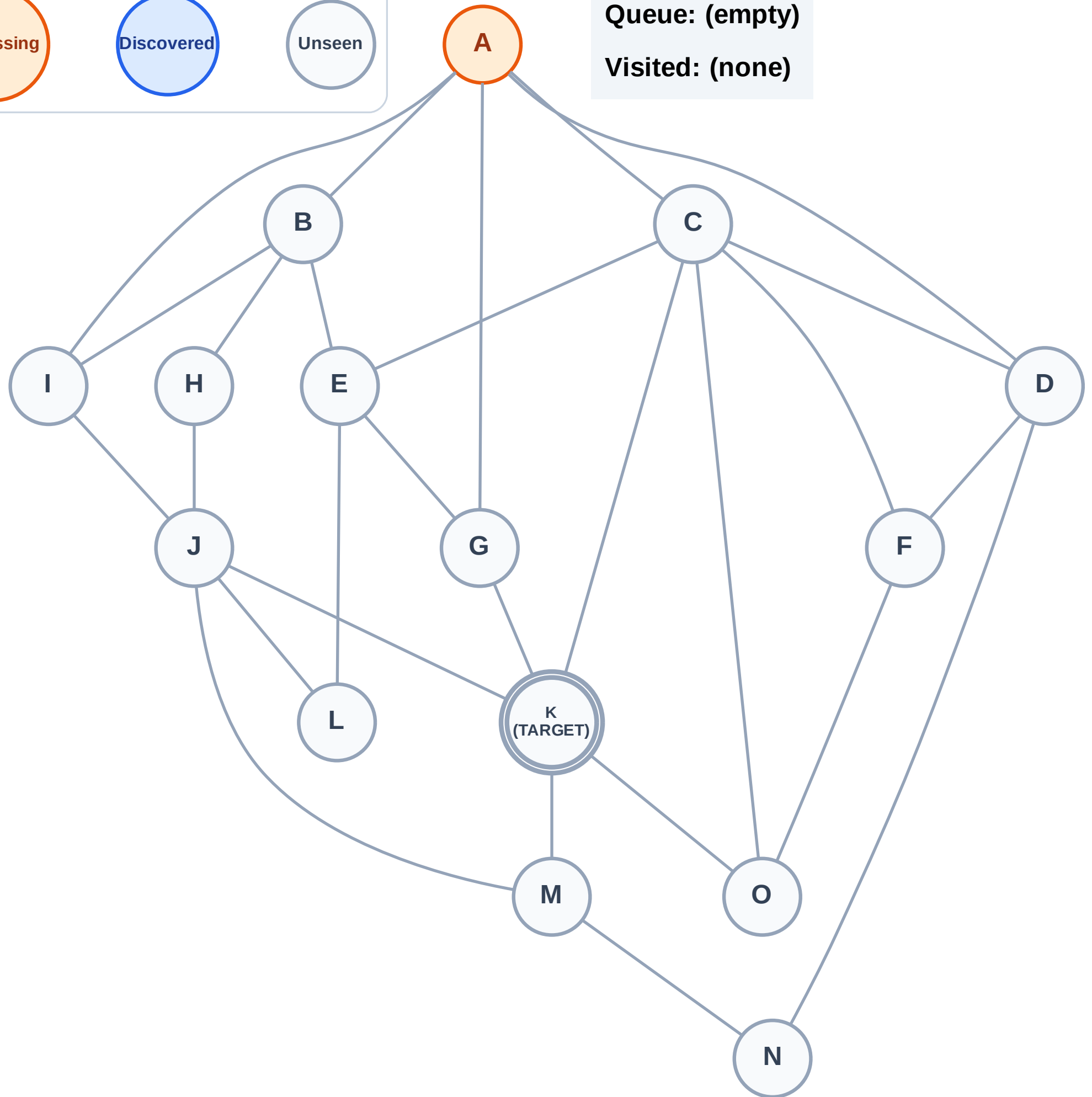
Step 2: Process node A

Legend



Queue: (empty)

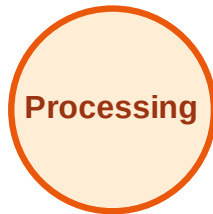
Visited: (none)



BFS Traversal

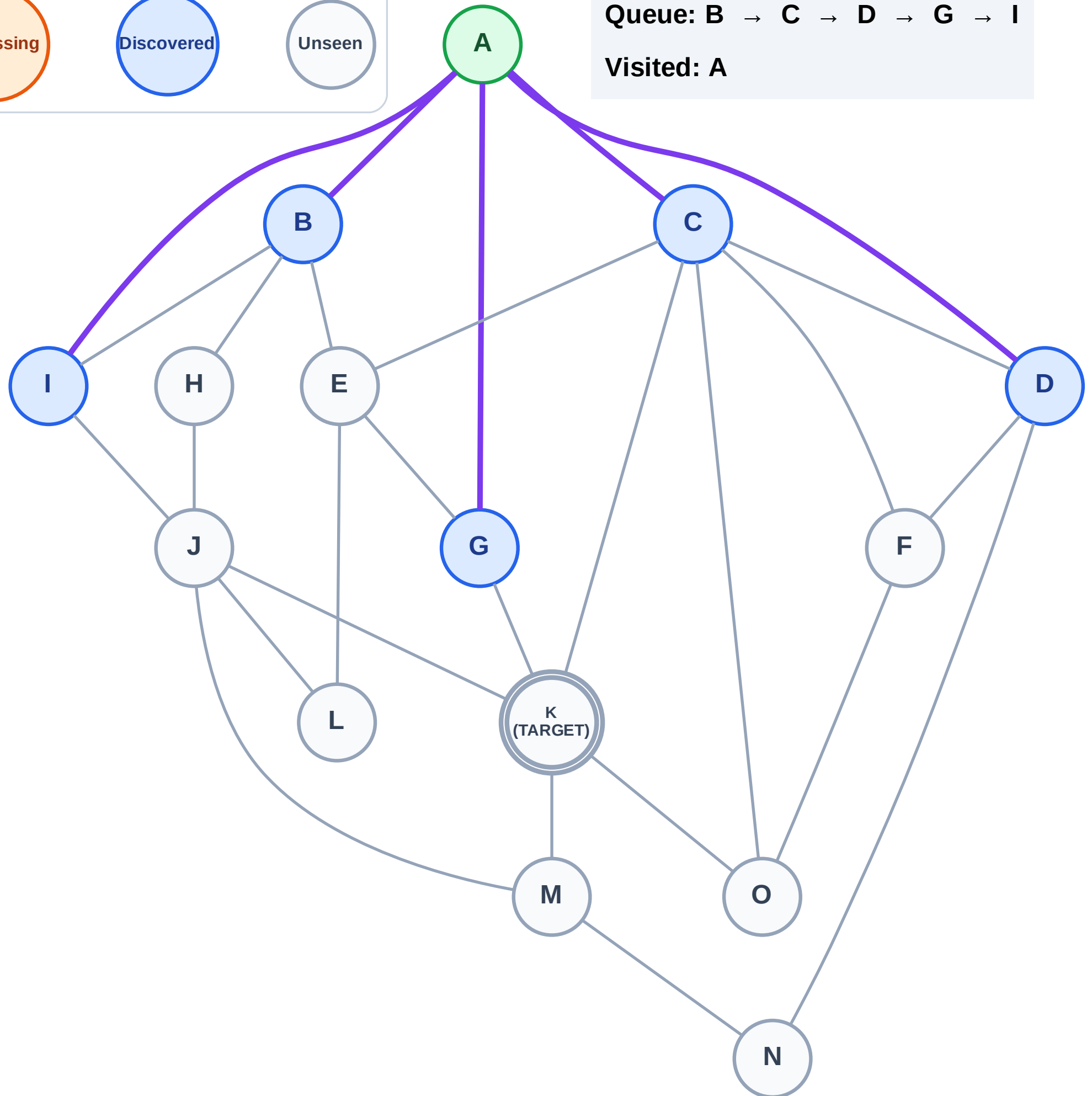
Step 3: Discover B, C, D, G, I from A

Legend



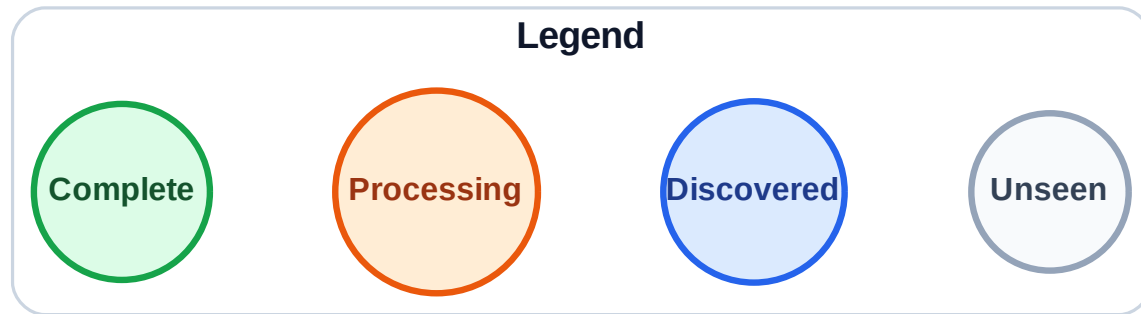
Queue: B → C → D → G → I

Visited: A



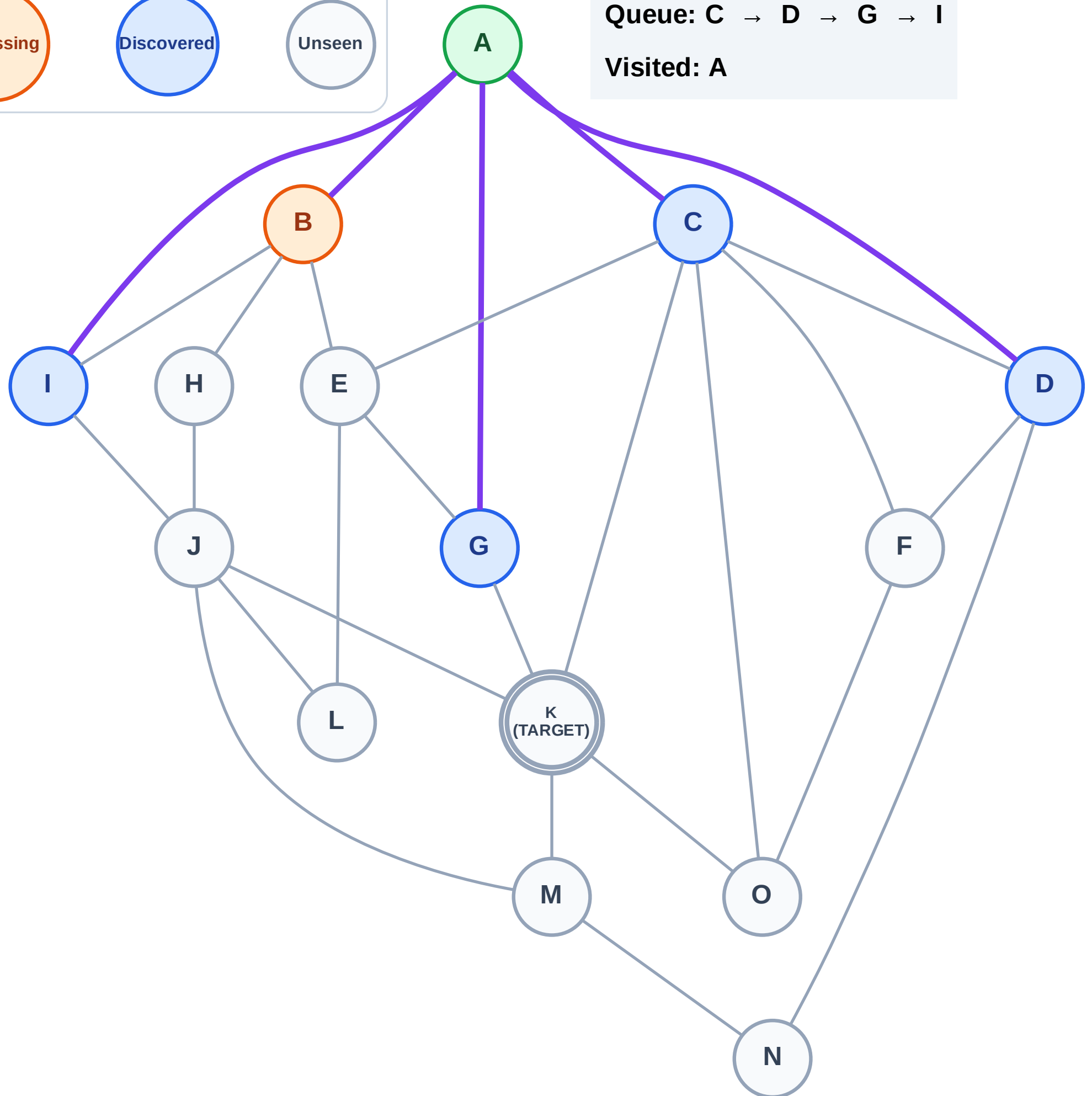
BFS Traversal

Step 4: Process node B



Queue: C → D → G → I

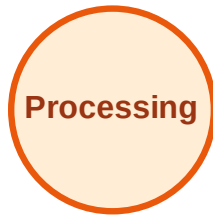
Visited: A



BFS Traversal

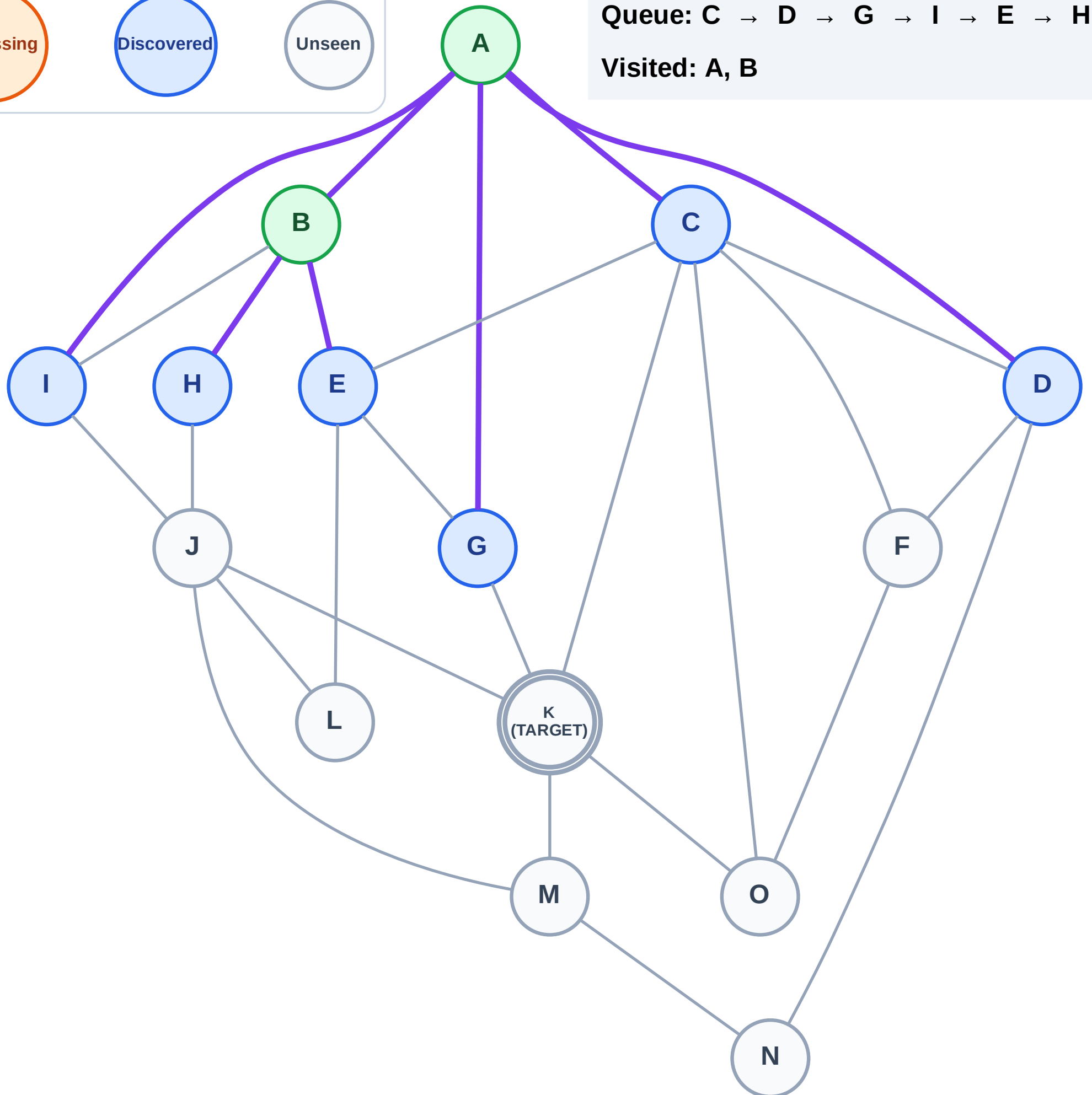
Step 5: Discover E, H from B

Legend



Queue: C → D → G → I → E → H

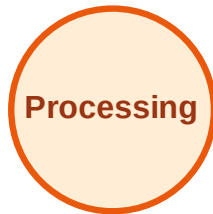
Visited: A, B



BFS Traversal

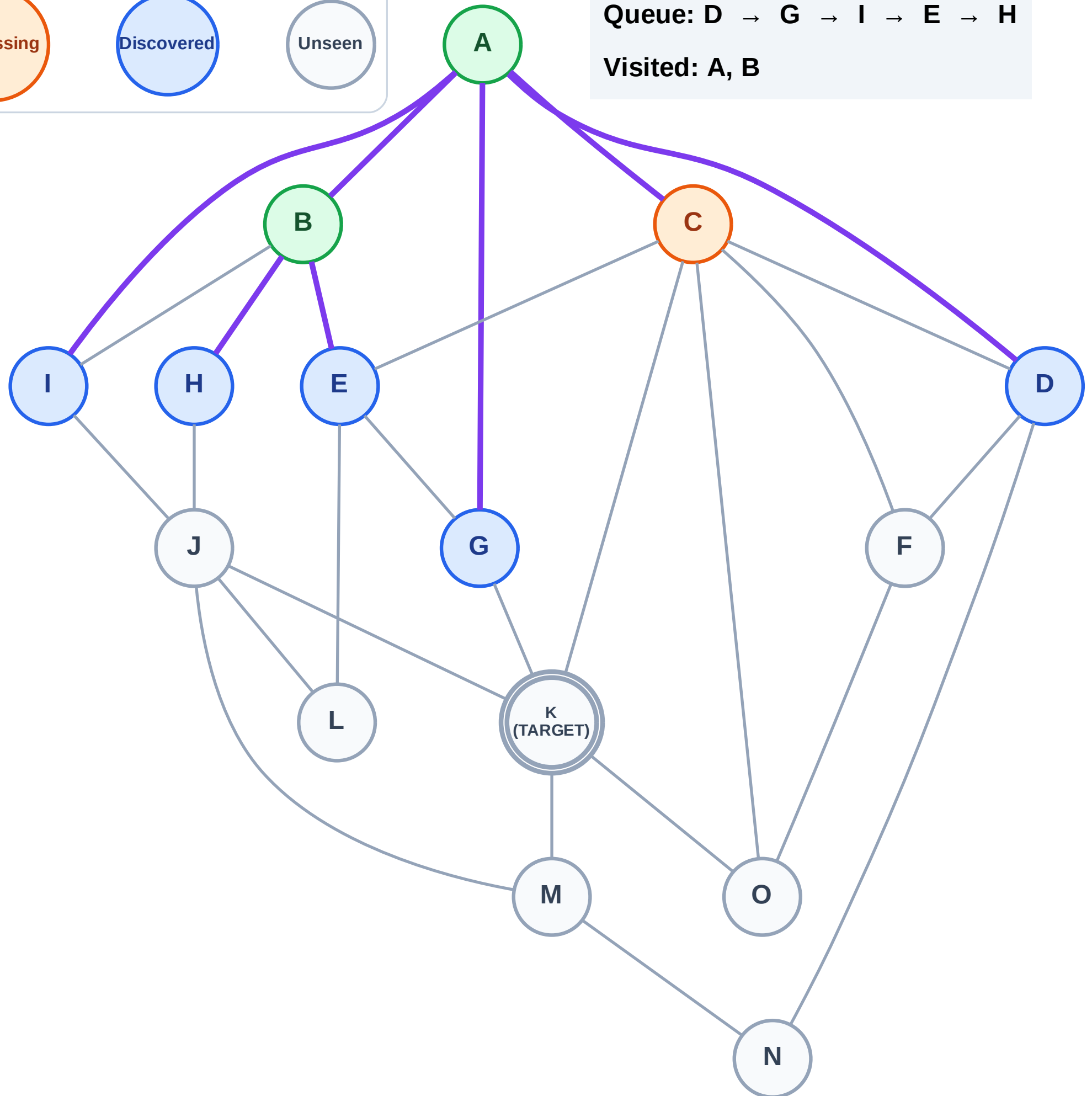
Step 6: Process node C

Legend



Queue: D → G → I → E → H

Visited: A, B



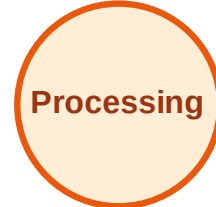
BFS Traversal

Step 7: Discover F, K, O from C

Legend



Complete



Processing



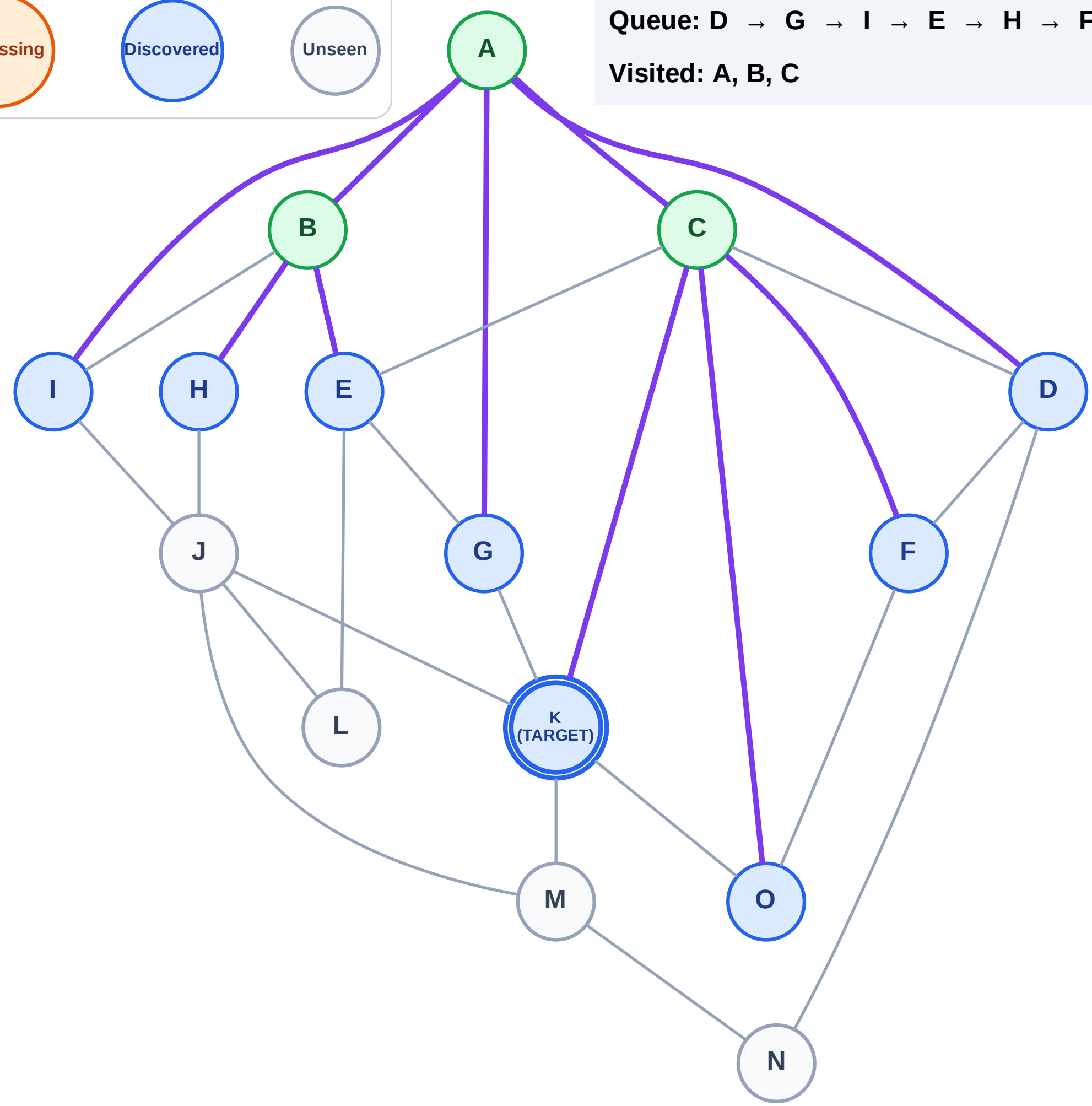
Discovered



Unseen

Queue: D → G → I → E → H → F → K → O

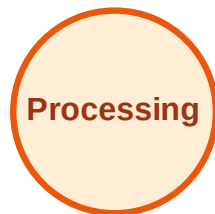
Visited: A, B, C



BFS Traversal

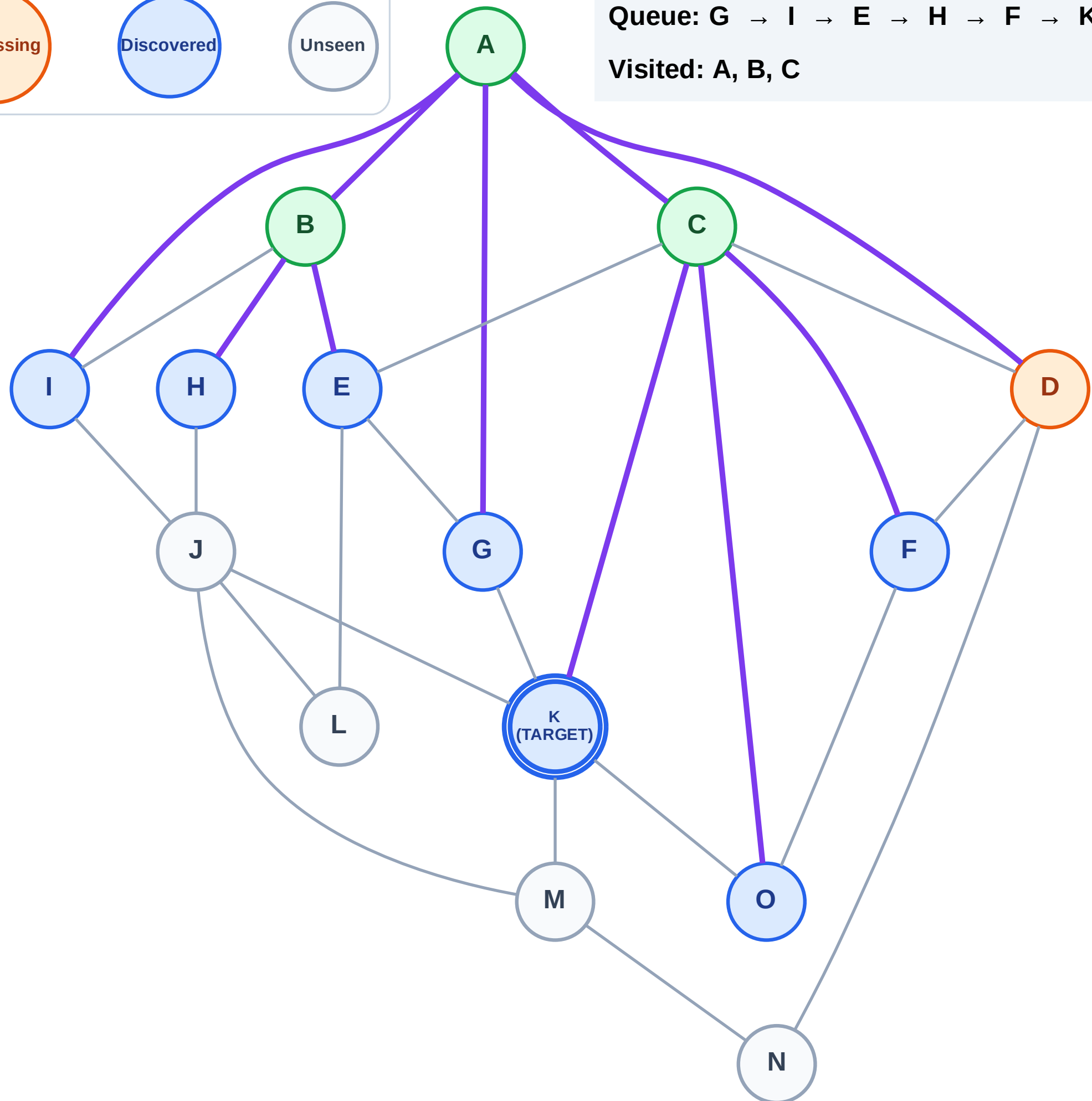
Step 8: Process node D

Legend



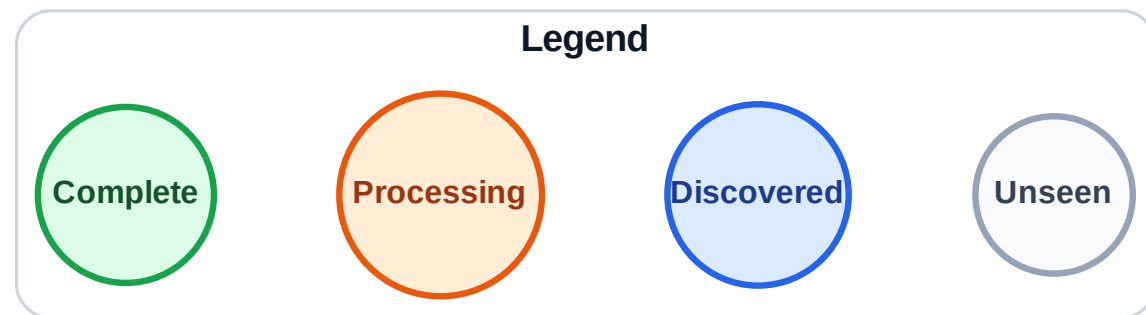
Queue: G → I → E → H → F → K → O

Visited: A, B, C



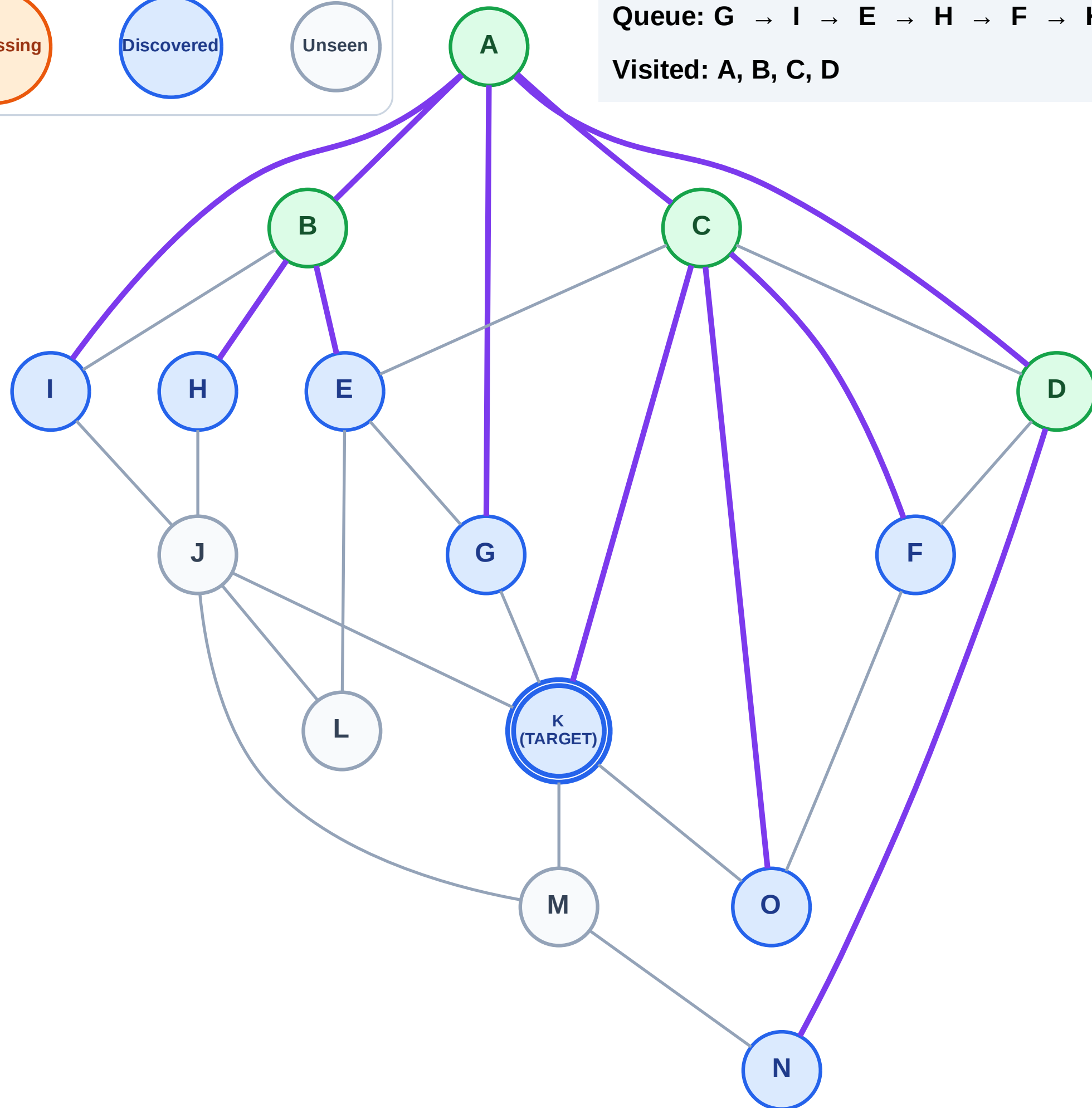
BFS Traversal

Step 9: Discover N from D



Queue: G → I → E → H → F → K → O → N

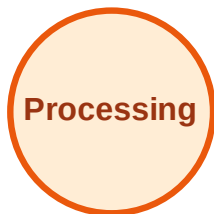
Visited: A, B, C, D



BFS Traversal

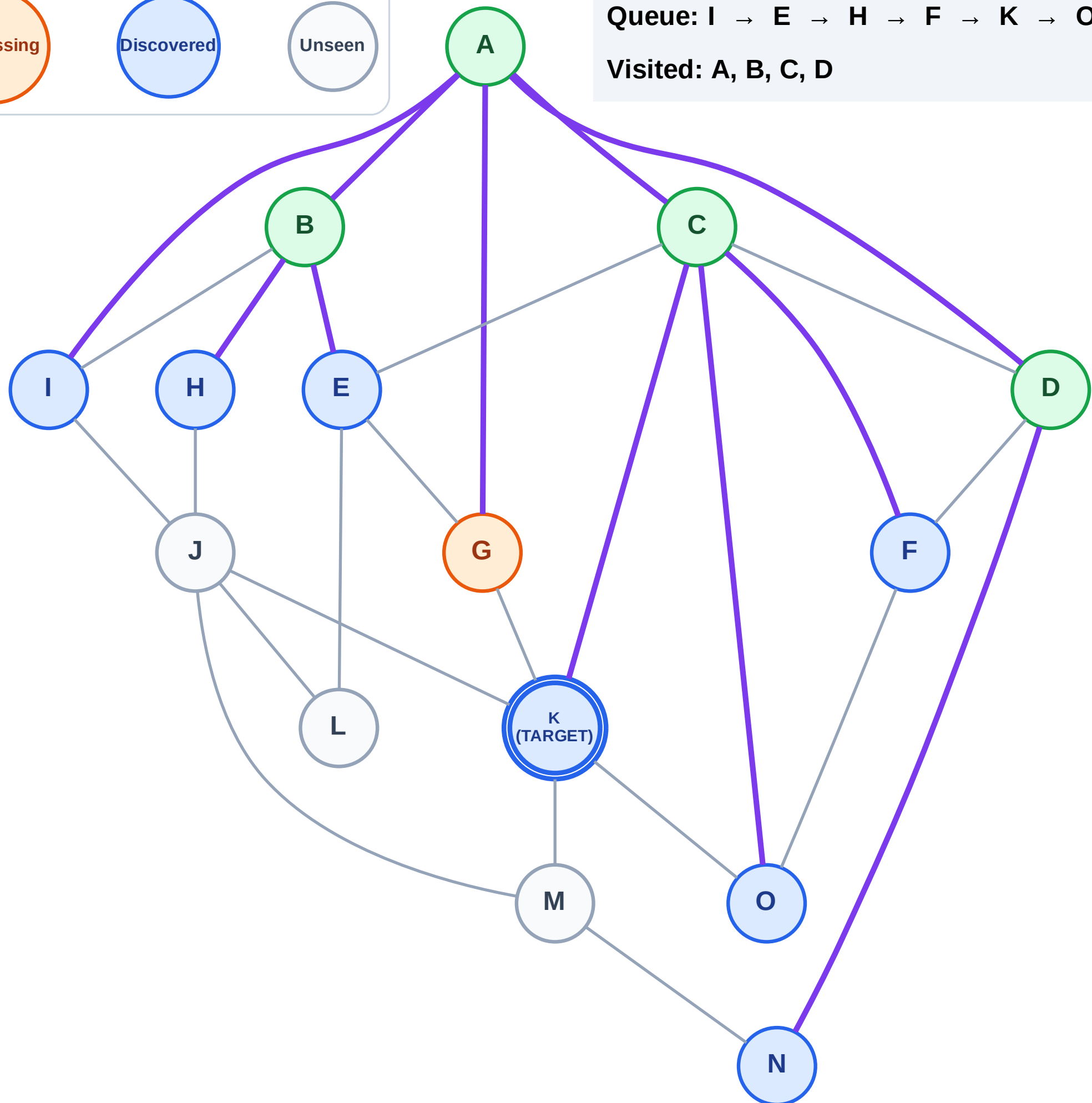
Step 10: Process node G

Legend



Queue: I → E → H → F → K → O → N

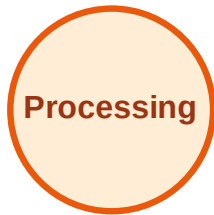
Visited: A, B, C, D



BFS Traversal

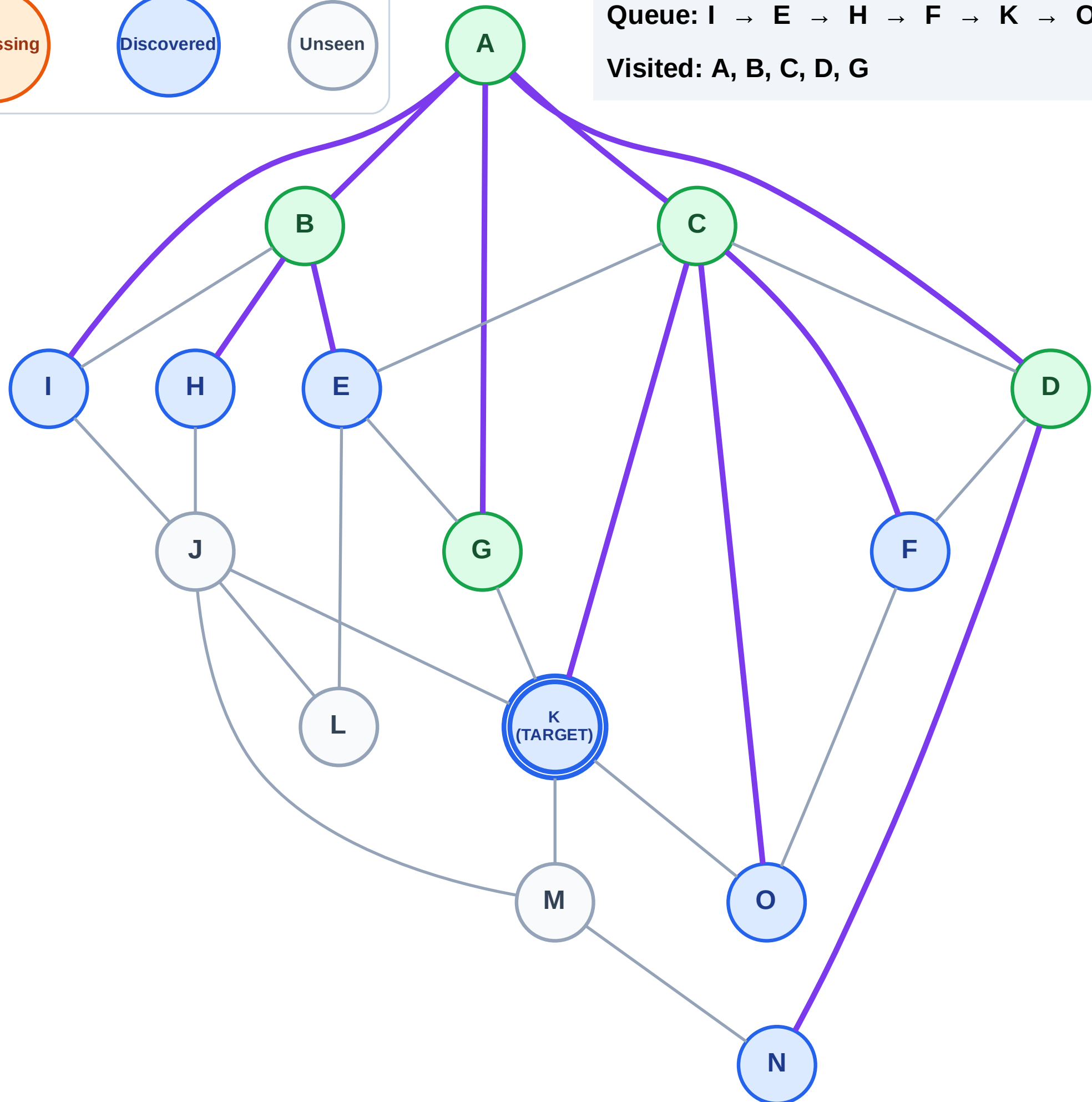
Step 11: No new neighbors from G

Legend



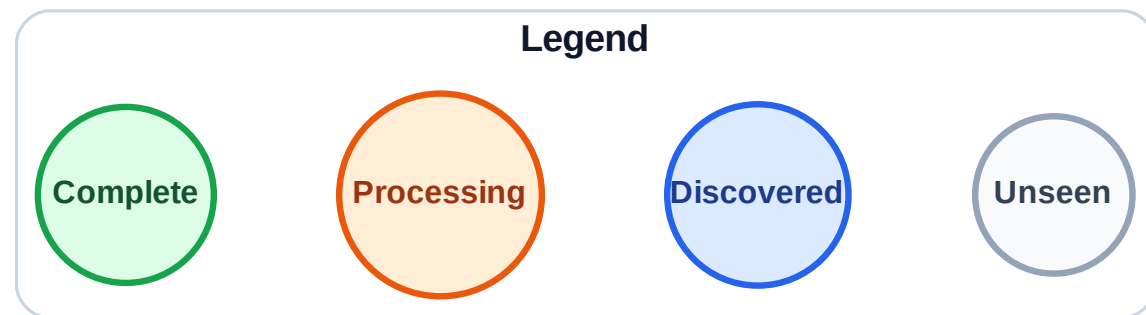
Queue: I → E → H → F → K → O → N

Visited: A, B, C, D, G



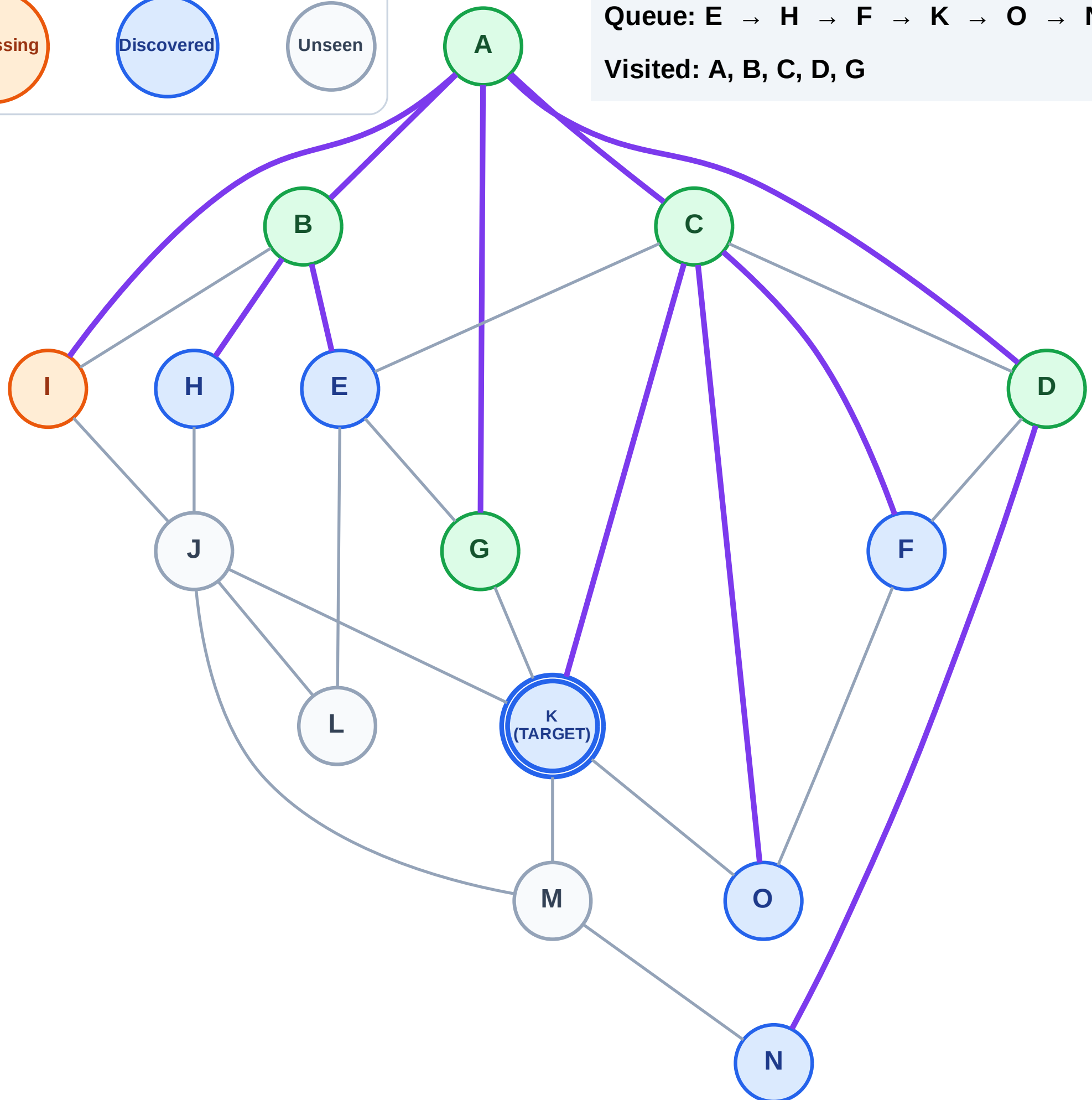
BFS Traversal

Step 12: Process node I



Queue: E → H → F → K → O → N

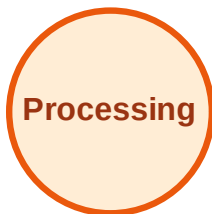
Visited: A, B, C, D, G



BFS Traversal

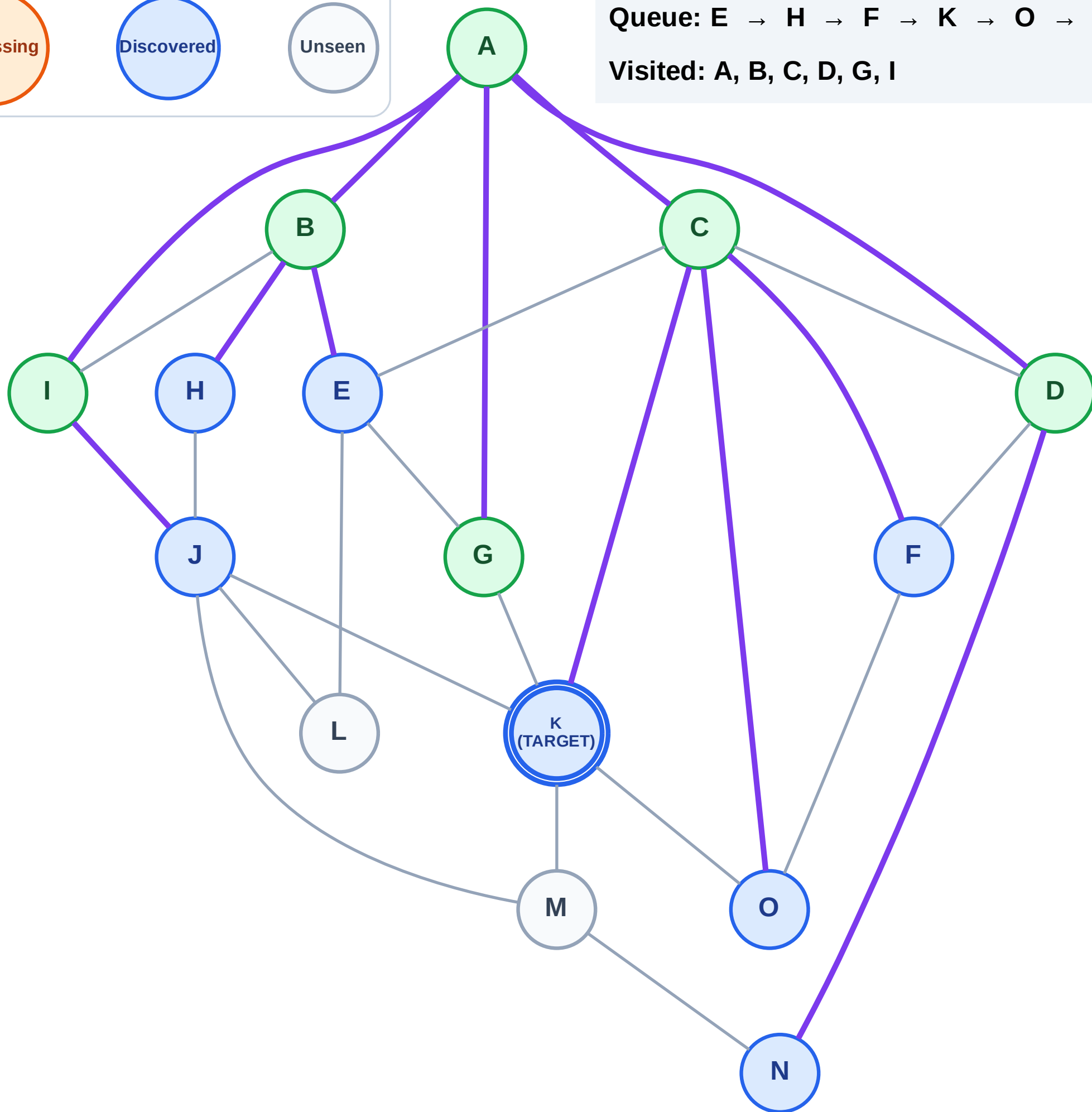
Step 13: Discover J from I

Legend



Queue: E → H → F → K → O → N → J

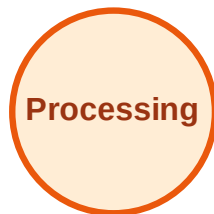
Visited: A, B, C, D, G, I



BFS Traversal

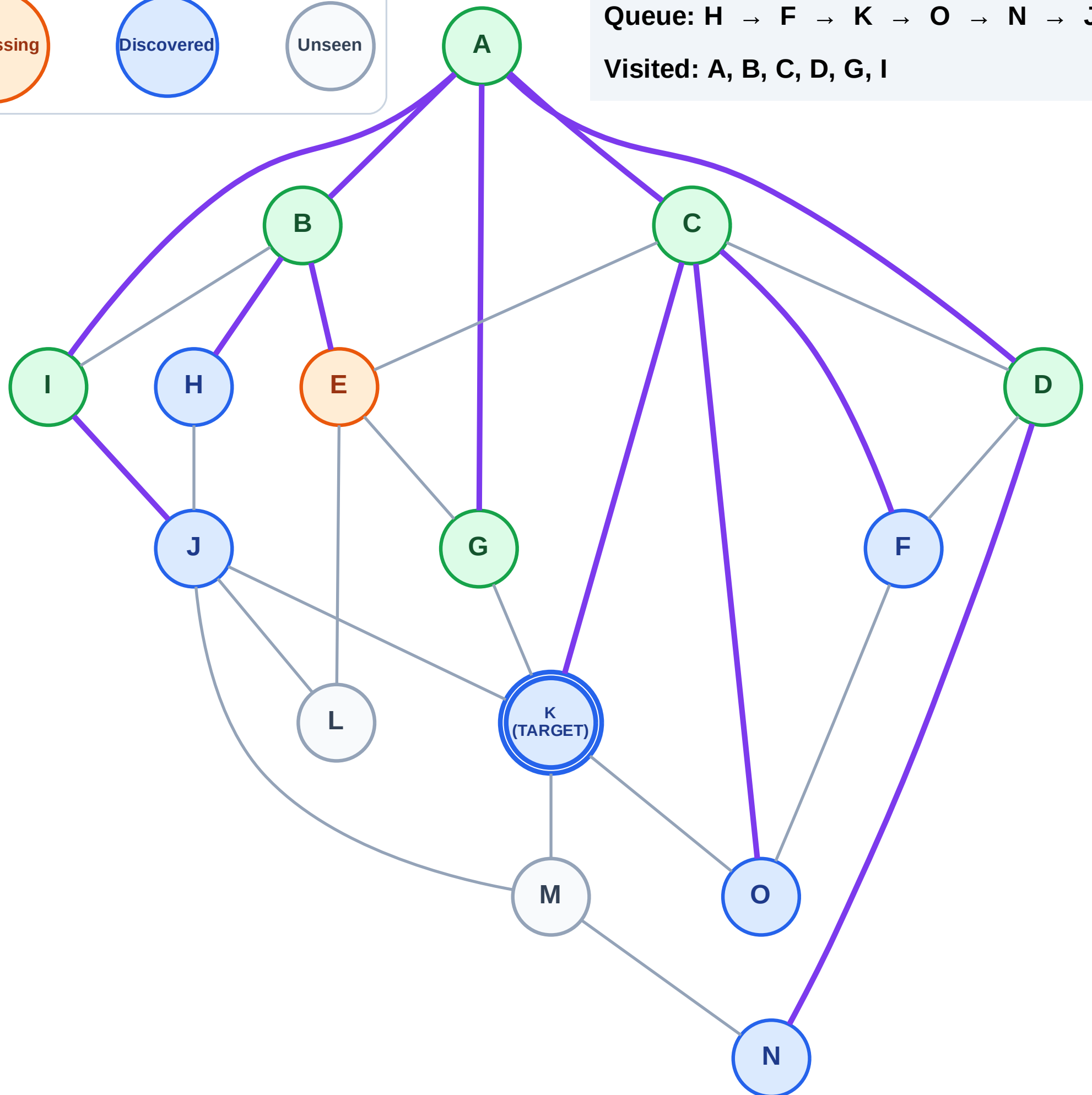
Step 14: Process node E

Legend



Queue: H → F → K → O → N → J

Visited: A, B, C, D, G, I



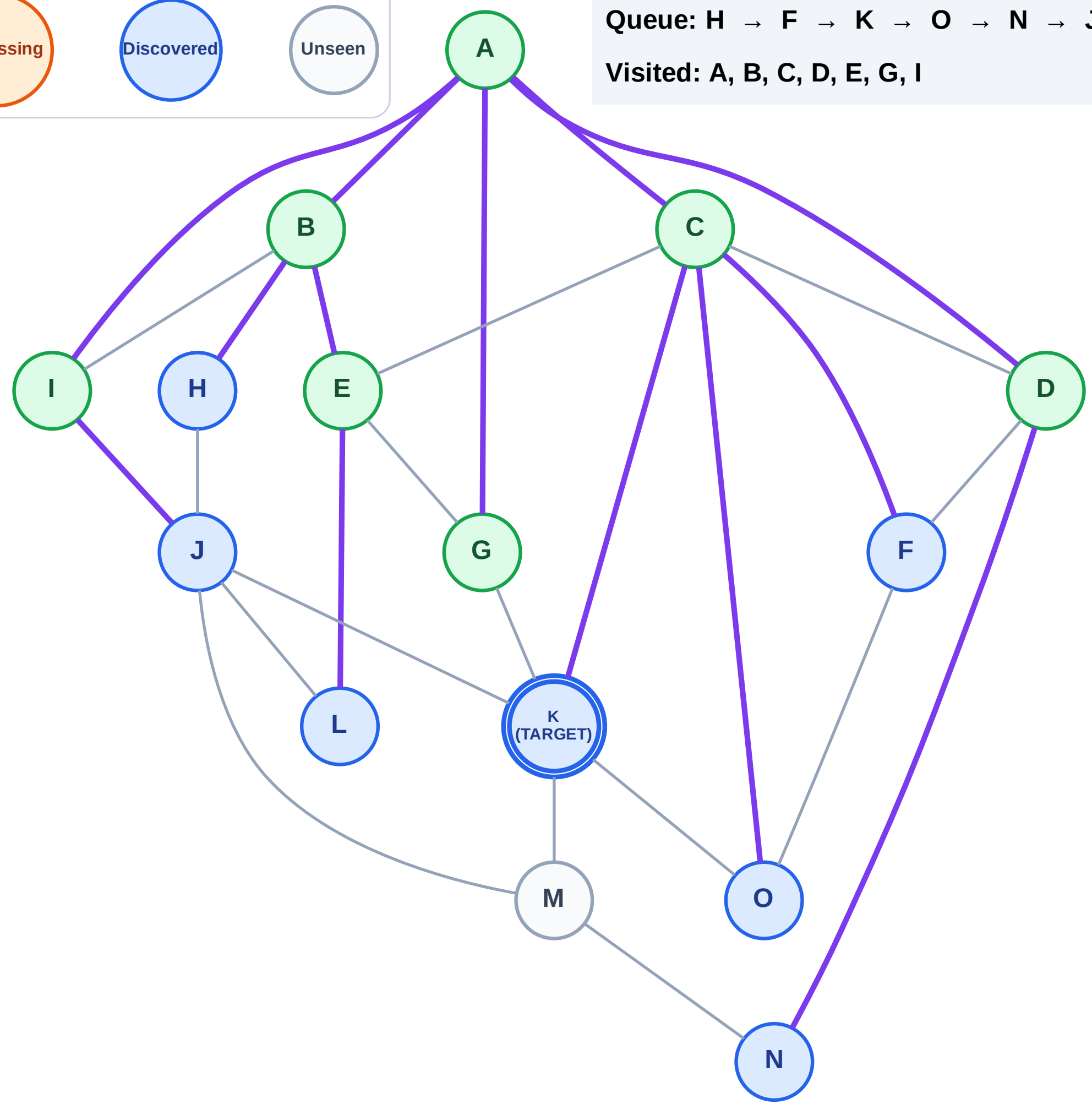
BFS Traversal

Step 15: Discover L from E

Legend



Queue: H → F → K → O → N → J → L
Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node H

Legend

Complete

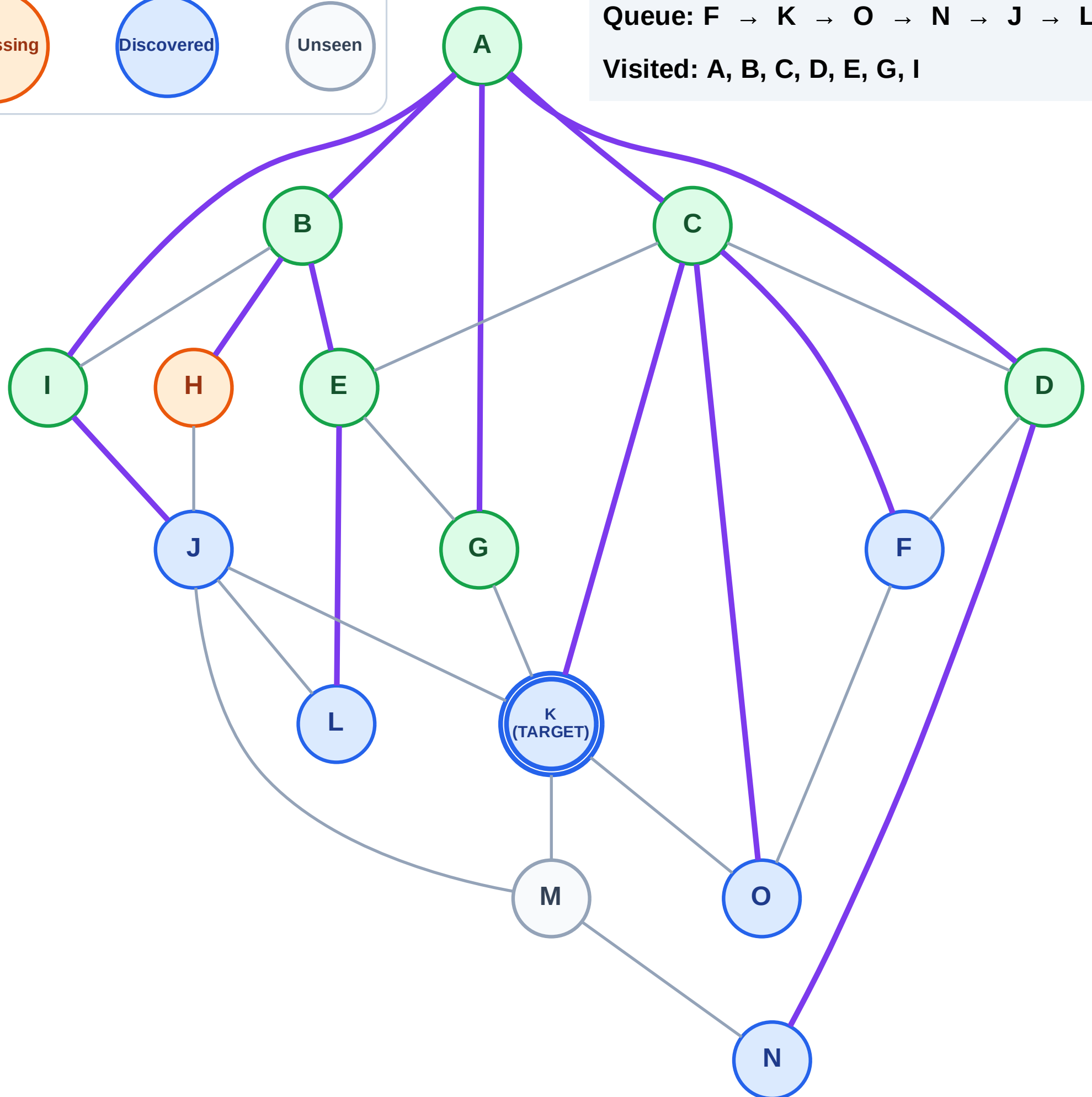
Processing

Discovered

Unseen

Queue: F → K → O → N → J → L

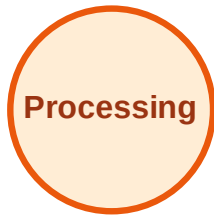
Visited: A, B, C, D, E, G, I



BFS Traversal

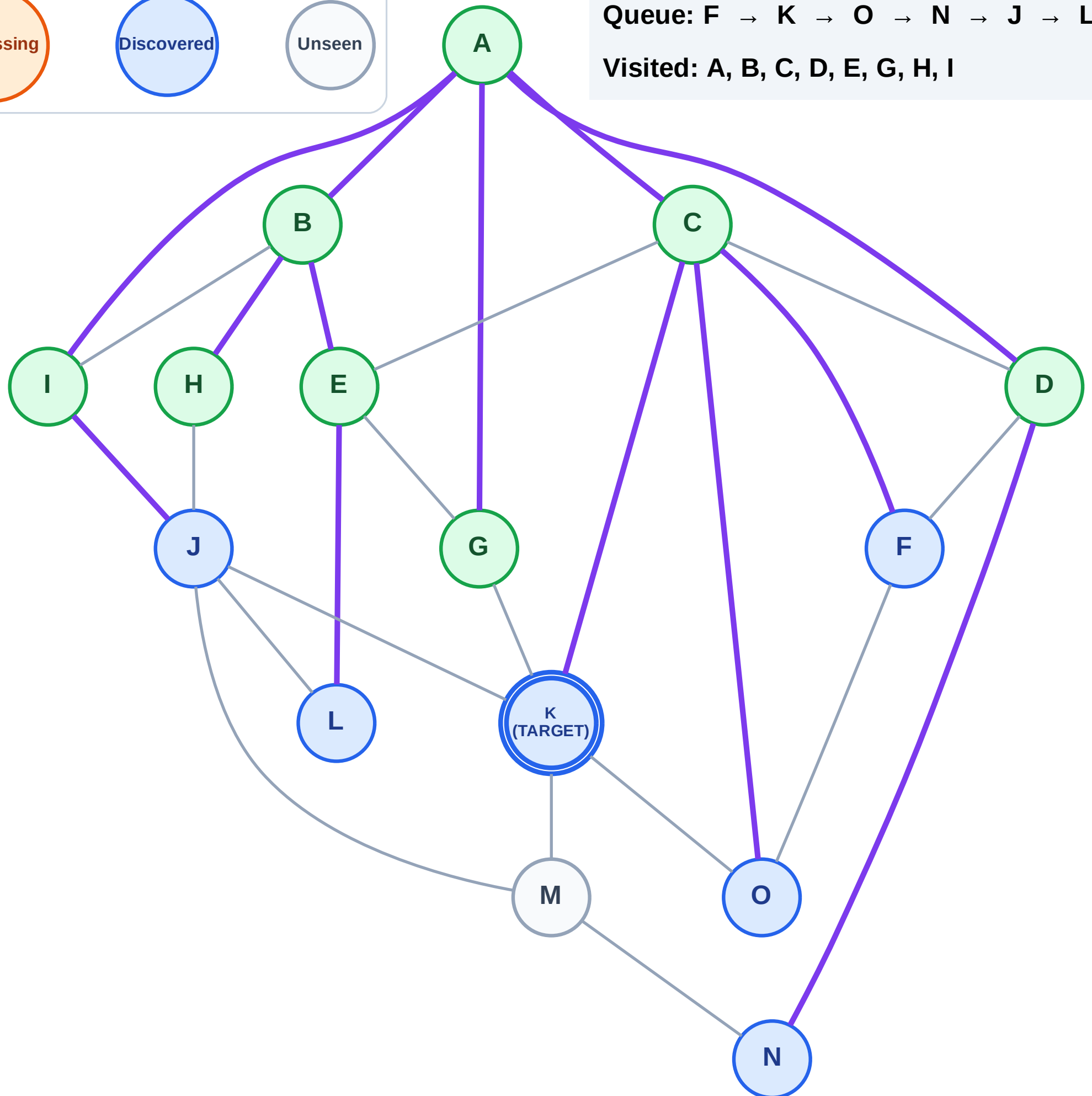
Step 17: No new neighbors from H

Legend



Queue: F → K → O → N → J → L

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node F

Legend

Complete

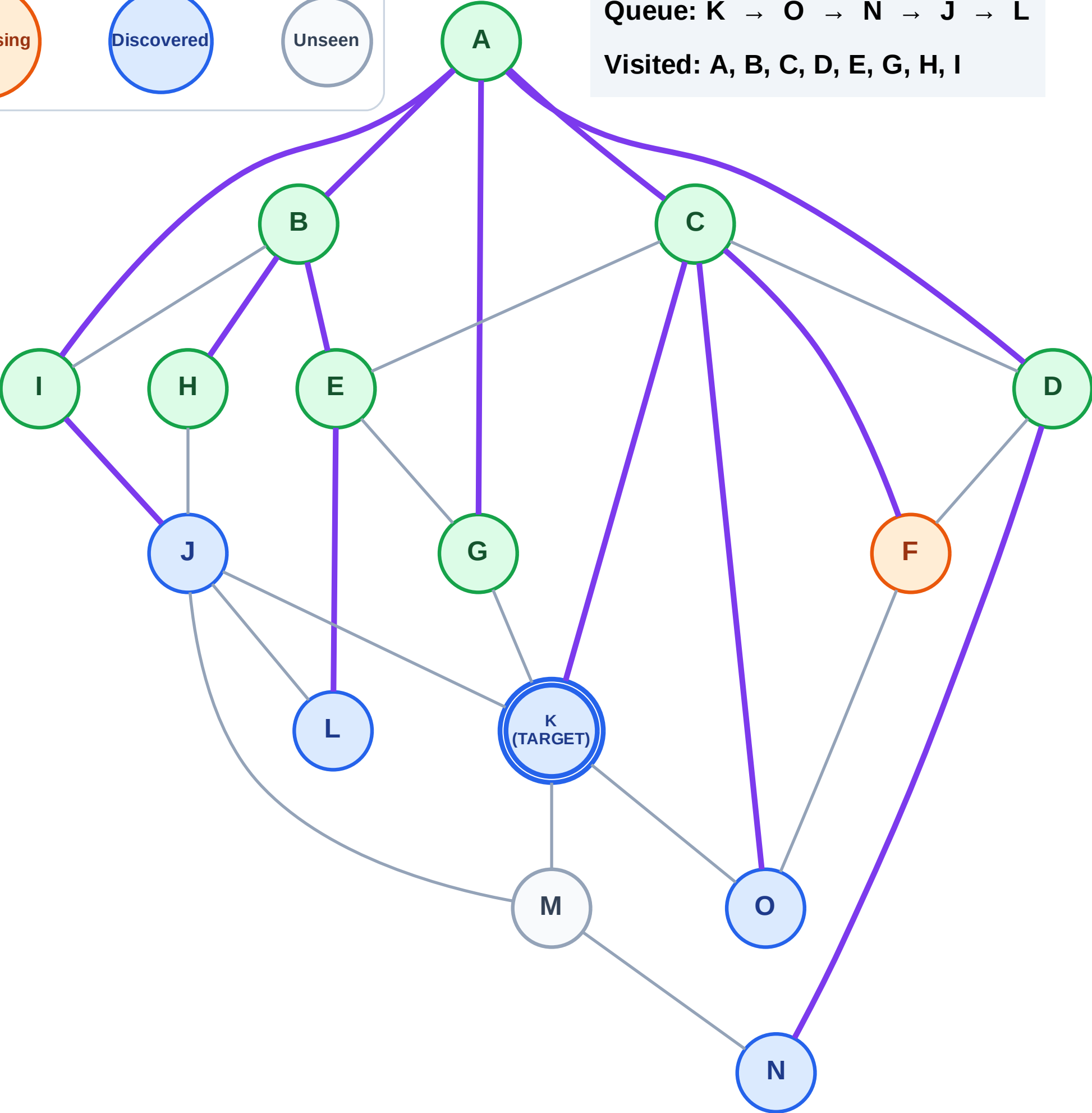
Processing

Discovered

Unseen

Queue: K → O → N → J → L

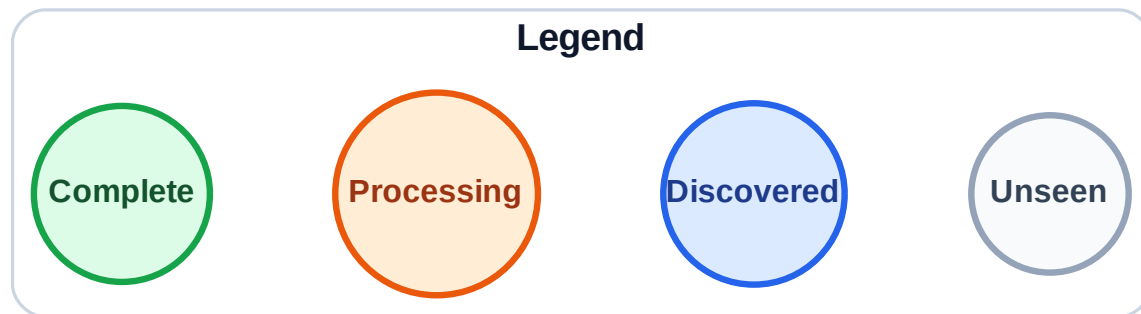
Visited: A, B, C, D, E, G, H, I



BFS Traversal

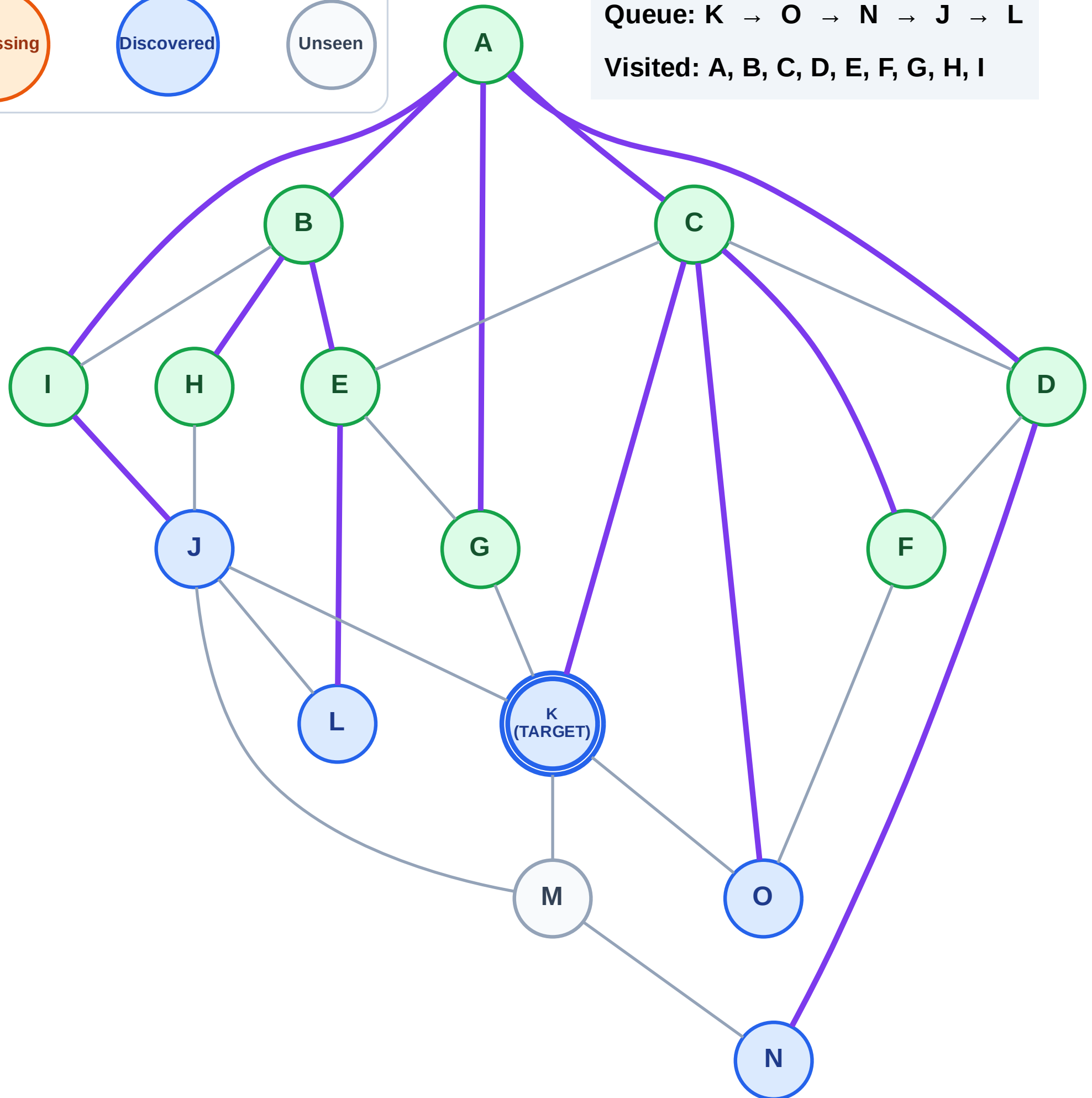
Step 19: No new neighbors from F

Step 19: No new neighbors from F



Queue: K → O → N → J → L

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node K

Legend

Complete

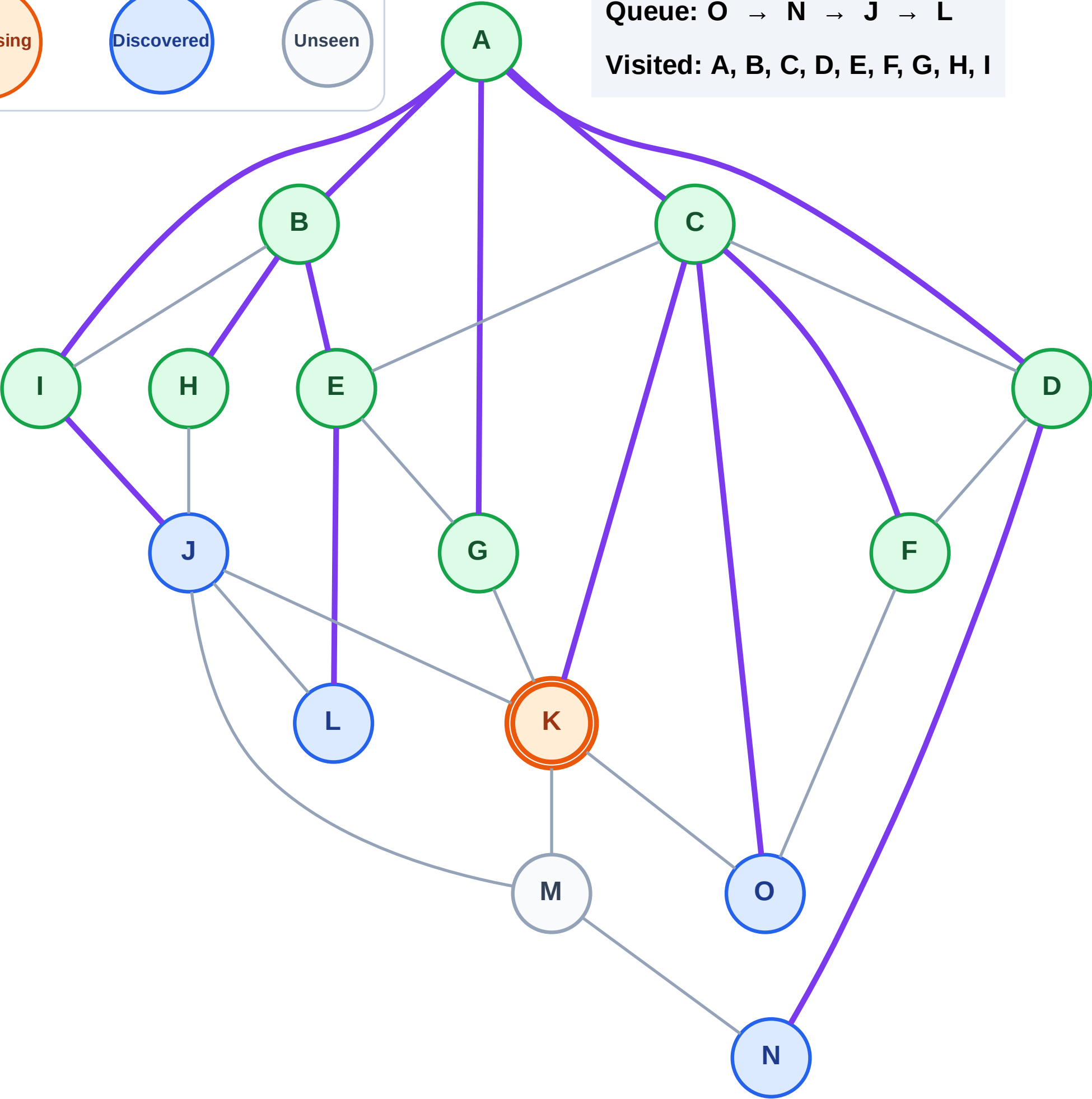
Processing

Discovered

Unseen

Queue: O → N → J → L

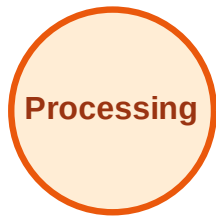
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

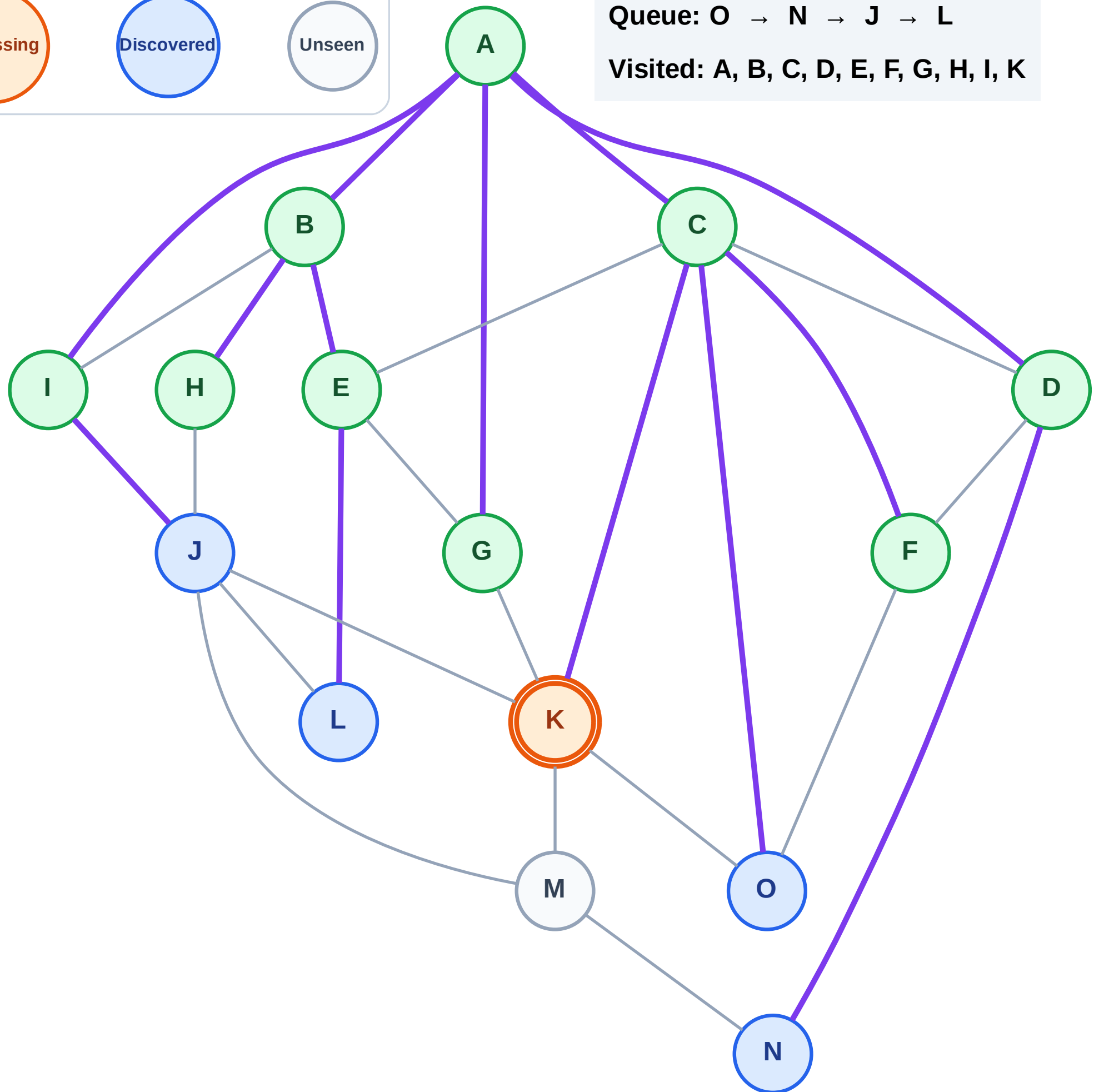
Step 21: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Queue: O → N → J → L

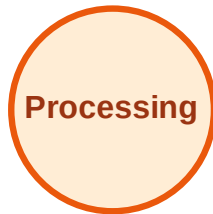
Visited: A, B, C, D, E, F, G, H, I, K



DFS Traversal

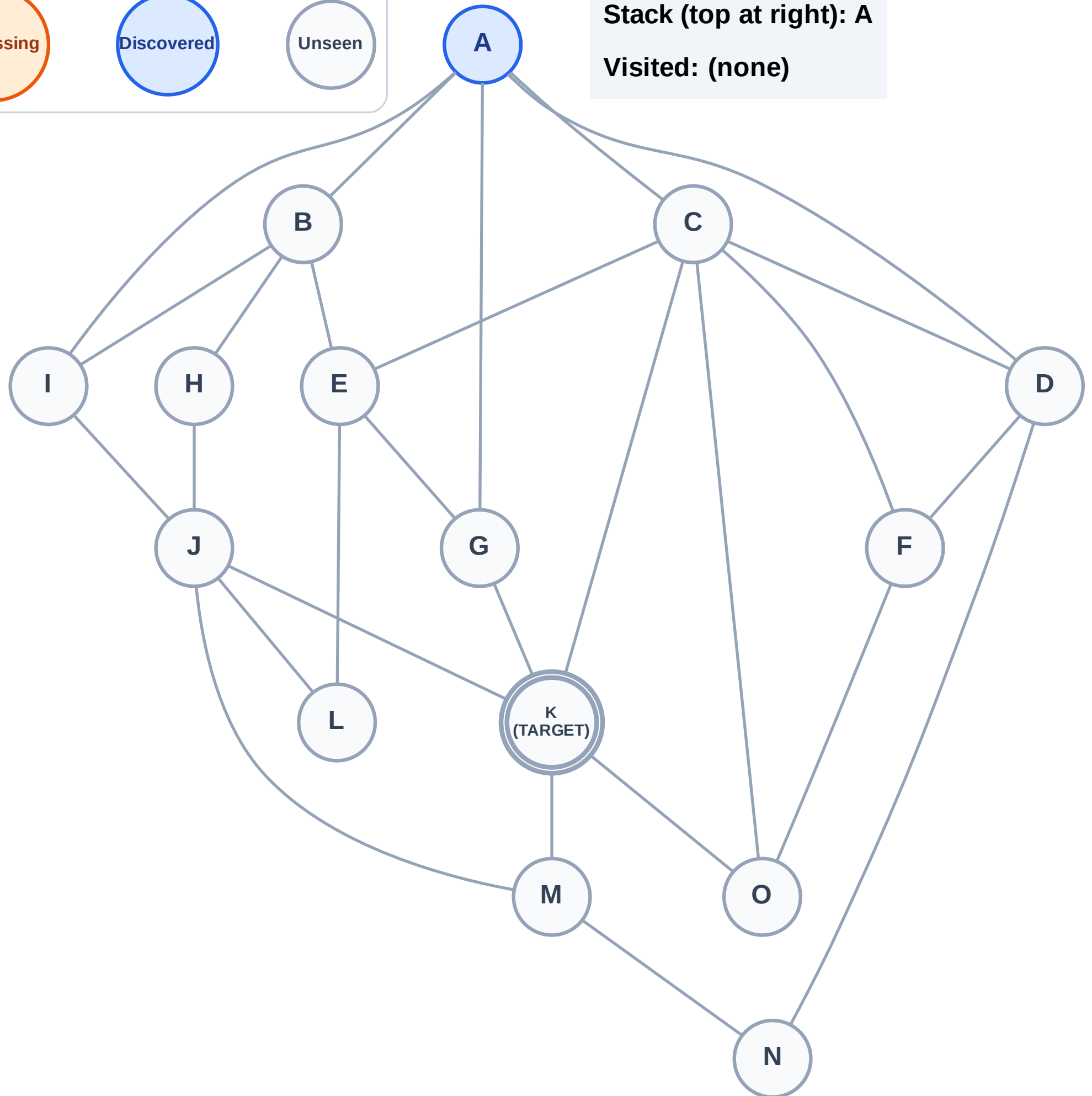
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

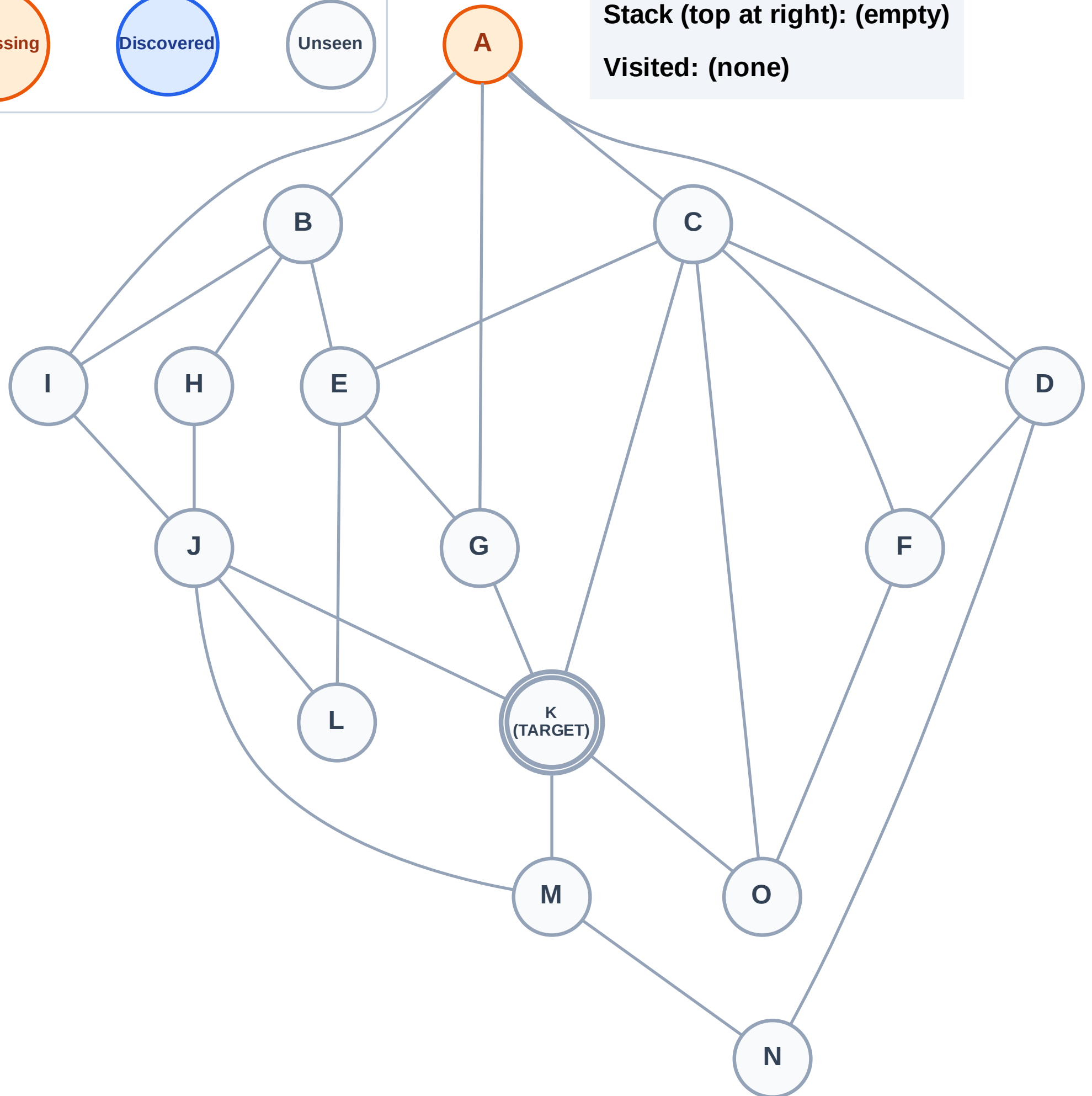
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover I, G, D, C, B from A

Legend

Complete

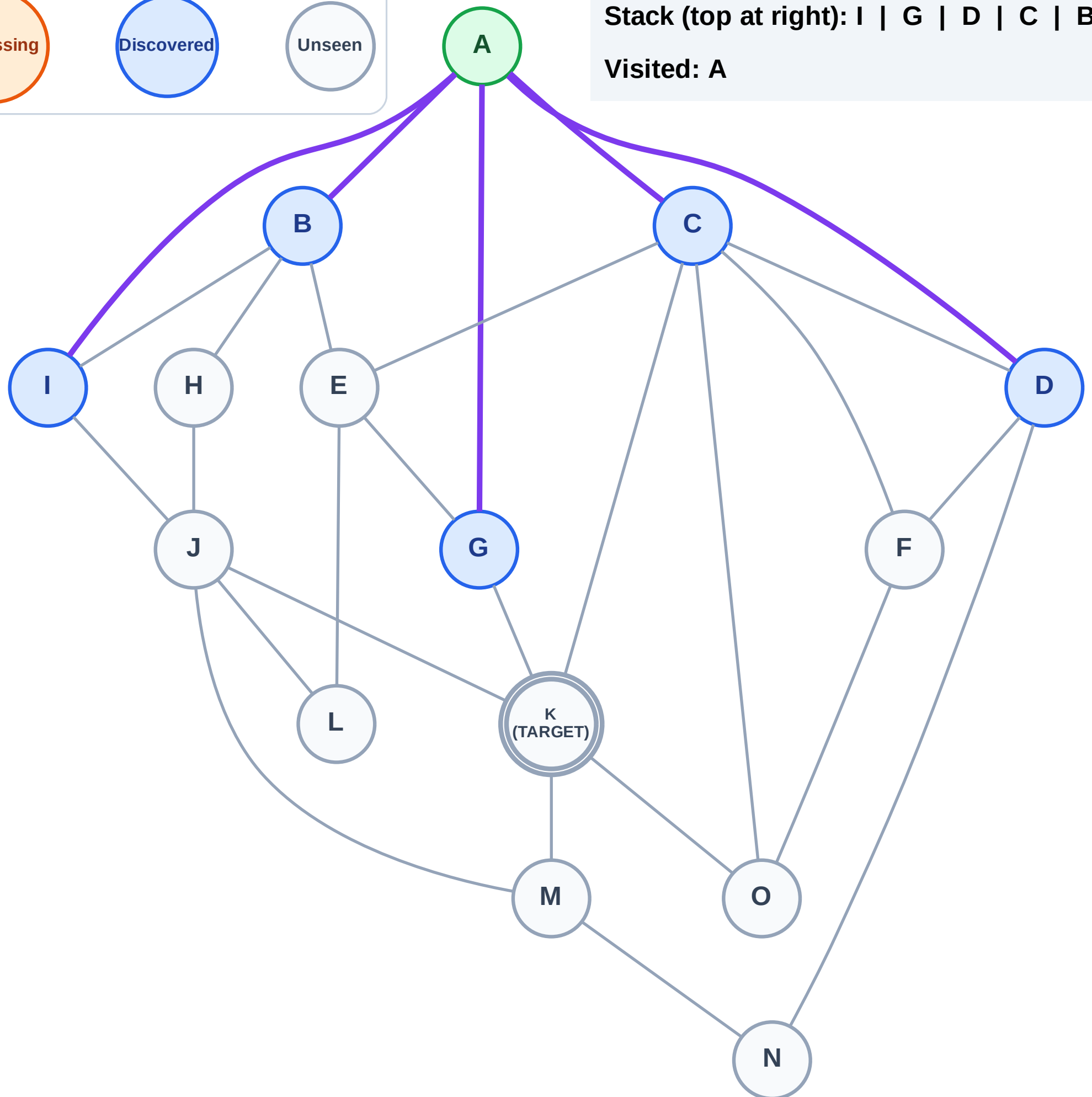
Processing

Discovered

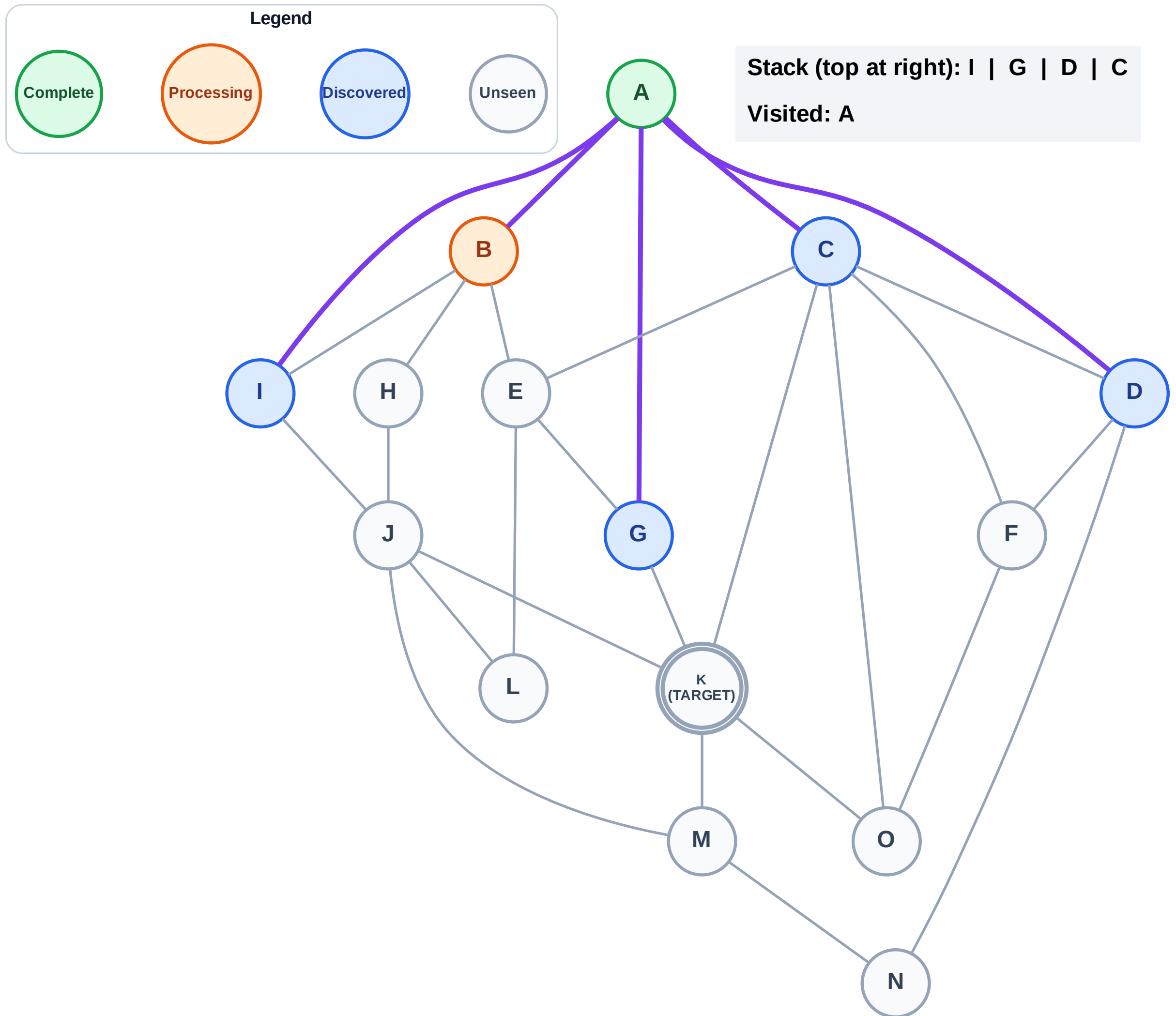
Unseen

Stack (top at right): I | G | D | C | B

Visited: A



Step 4: Process node B



DFS Traversal

Step 5: Discover H, E from B

Legend

Complete

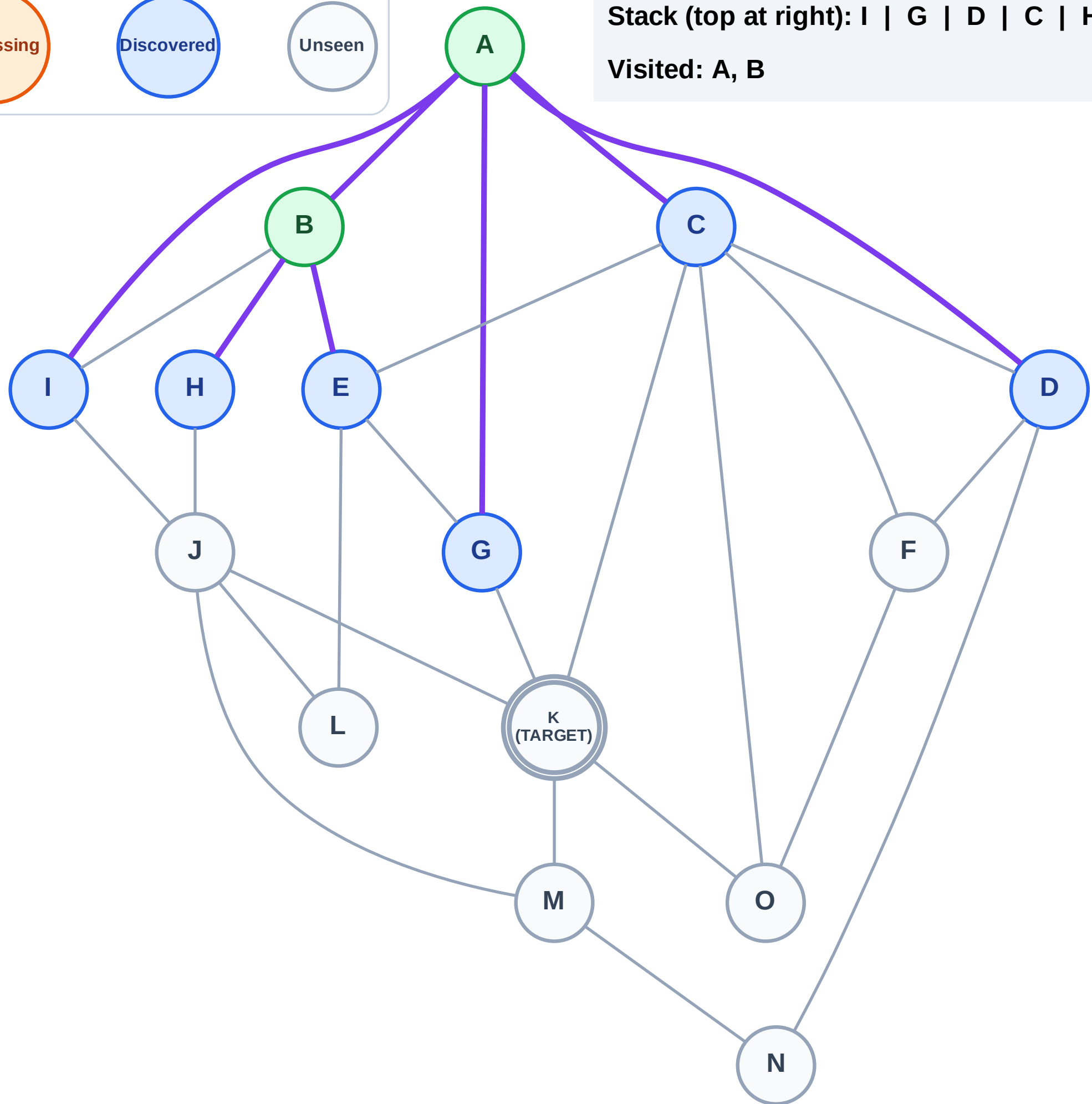
Processing

Discovered

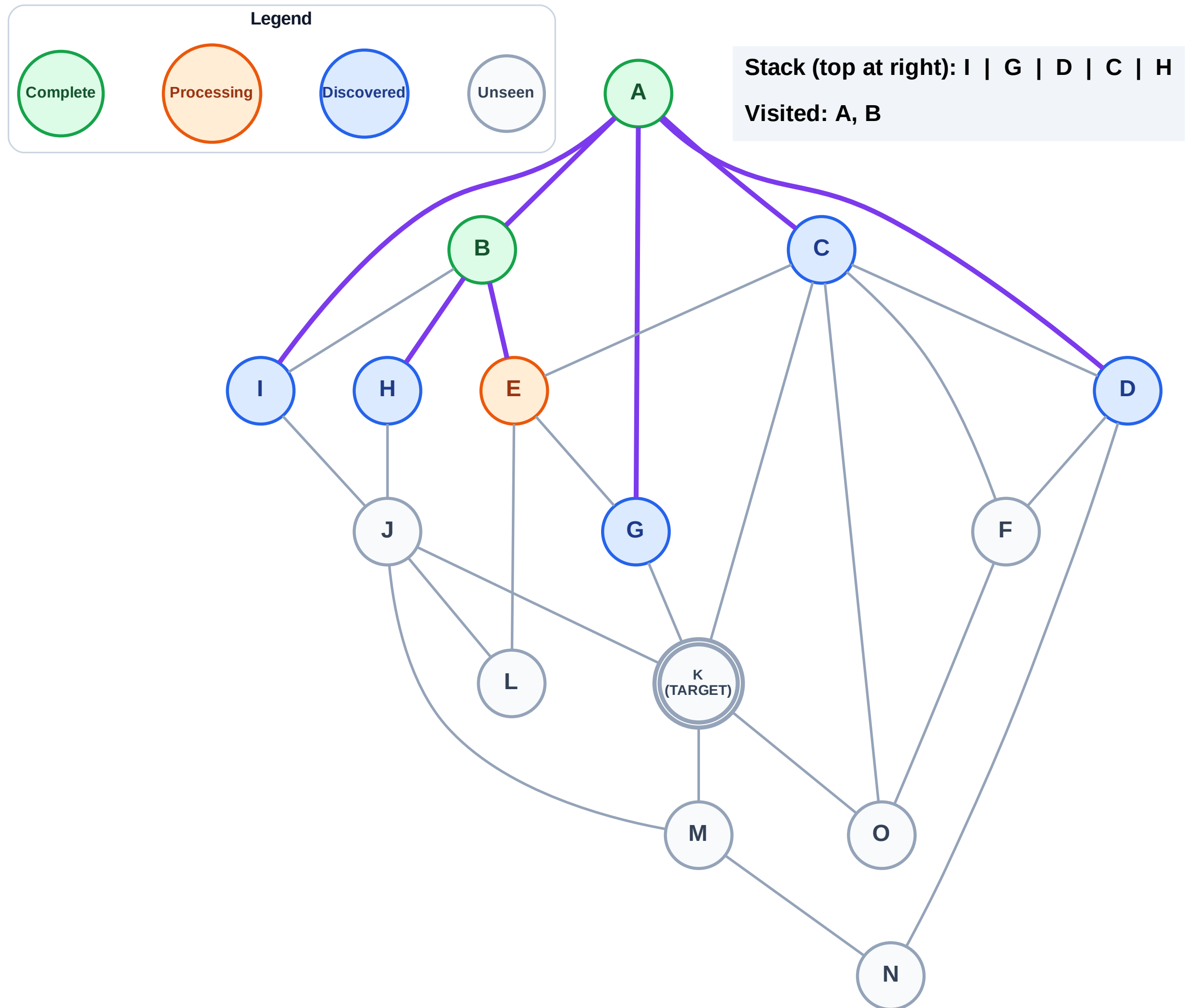
Unseen

Stack (top at right): I | G | D | C | H | E

Visited: A, B



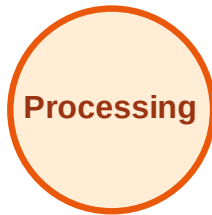
Step 6: Process node E



DFS Traversal

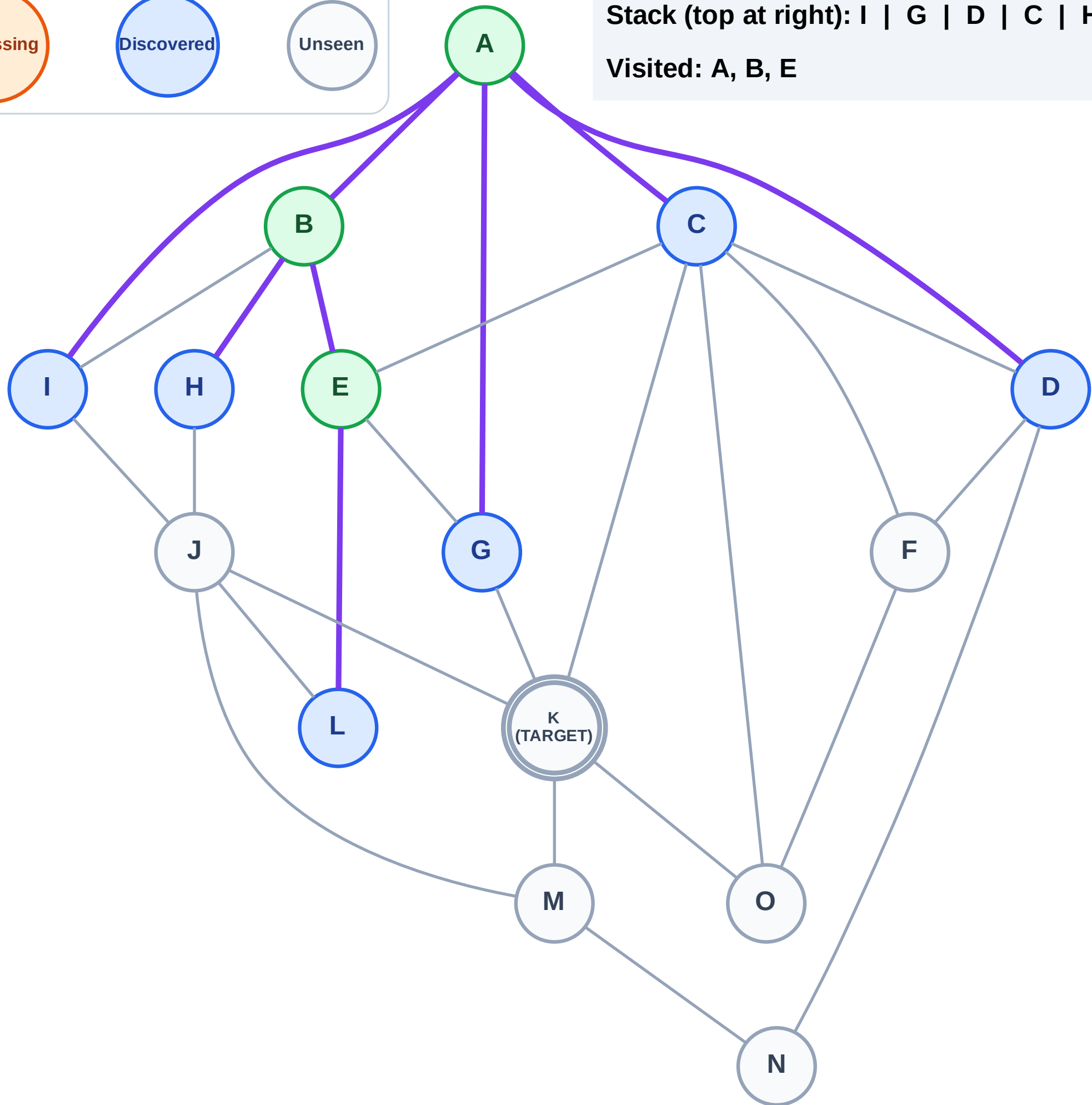
Step 7: Discover L from E

Legend



Stack (top at right): I | G | D | C | H | L

Visited: A, B, E



DFS Traversal

Step 8: Process node L

Legend

Complete

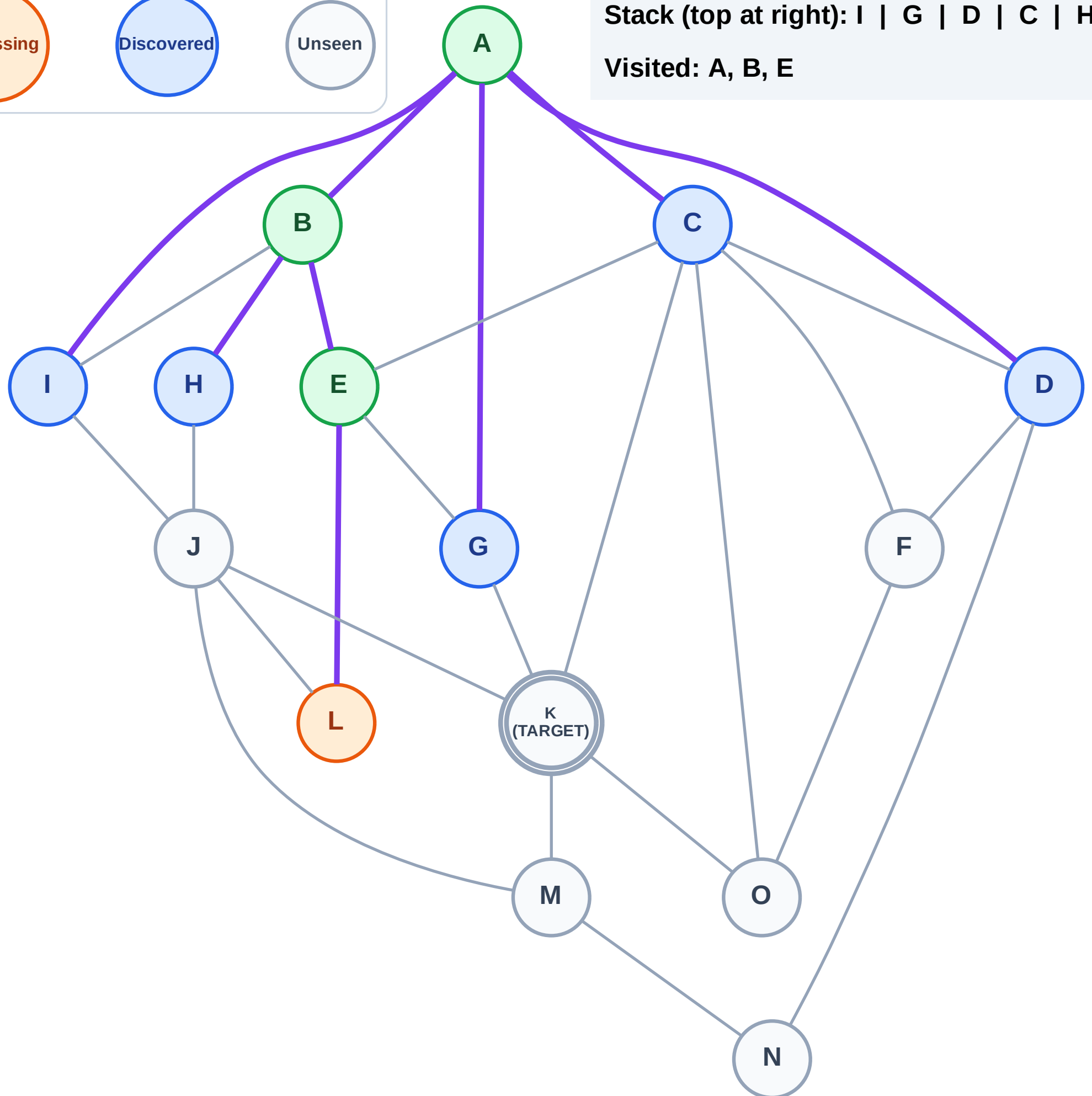
Processing

Discovered

Unseen

Stack (top at right): I | G | D | C | H

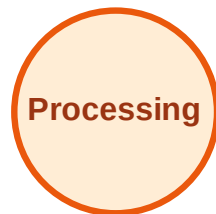
Visited: A, B, E



DFS Traversal

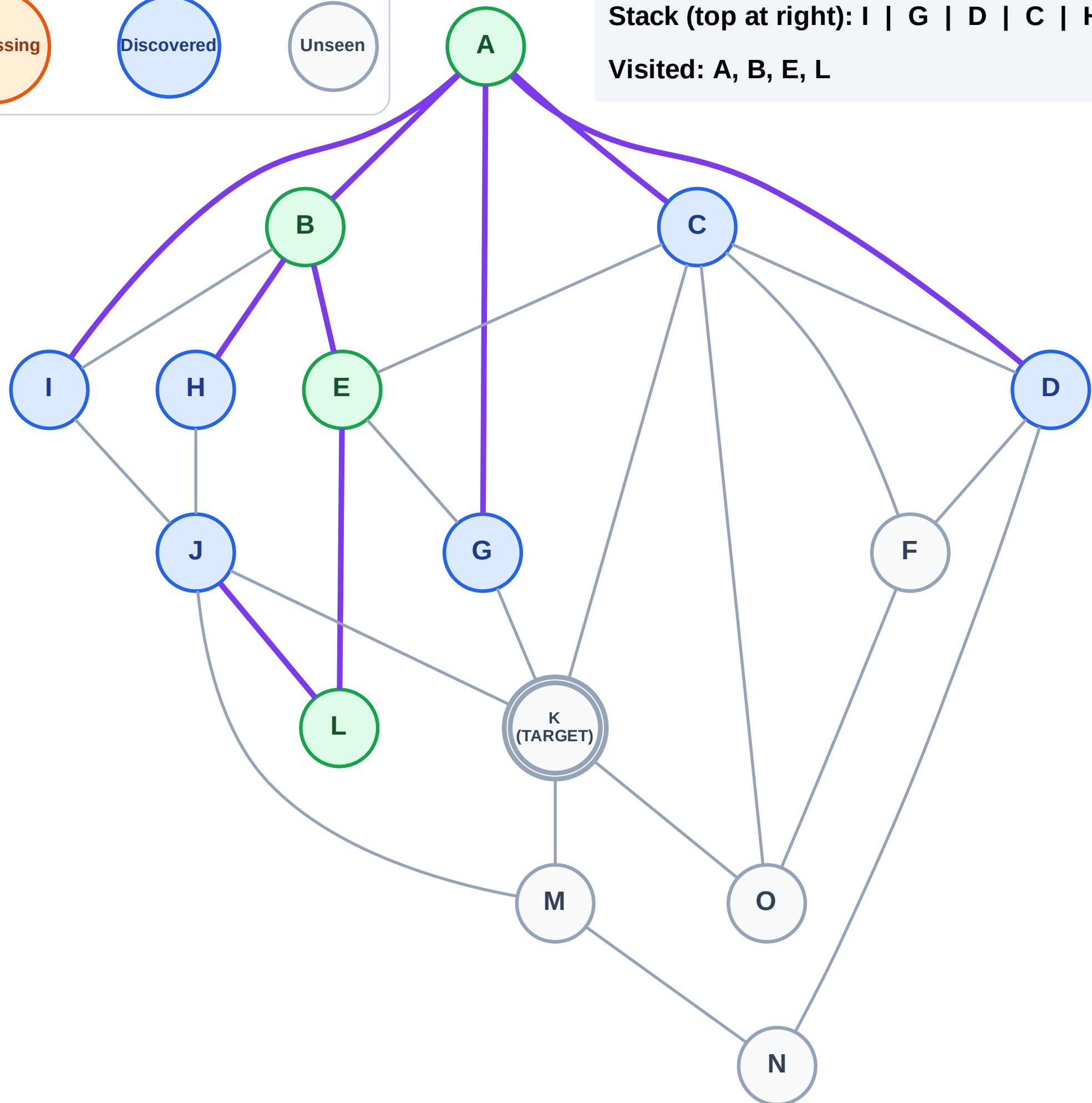
Step 9: Discover J from L

Legend



Stack (top at right): I | G | D | C | H | J

Visited: A, B, E, L



DFS Traversal

Step 10: Process node J

Legend

Complete

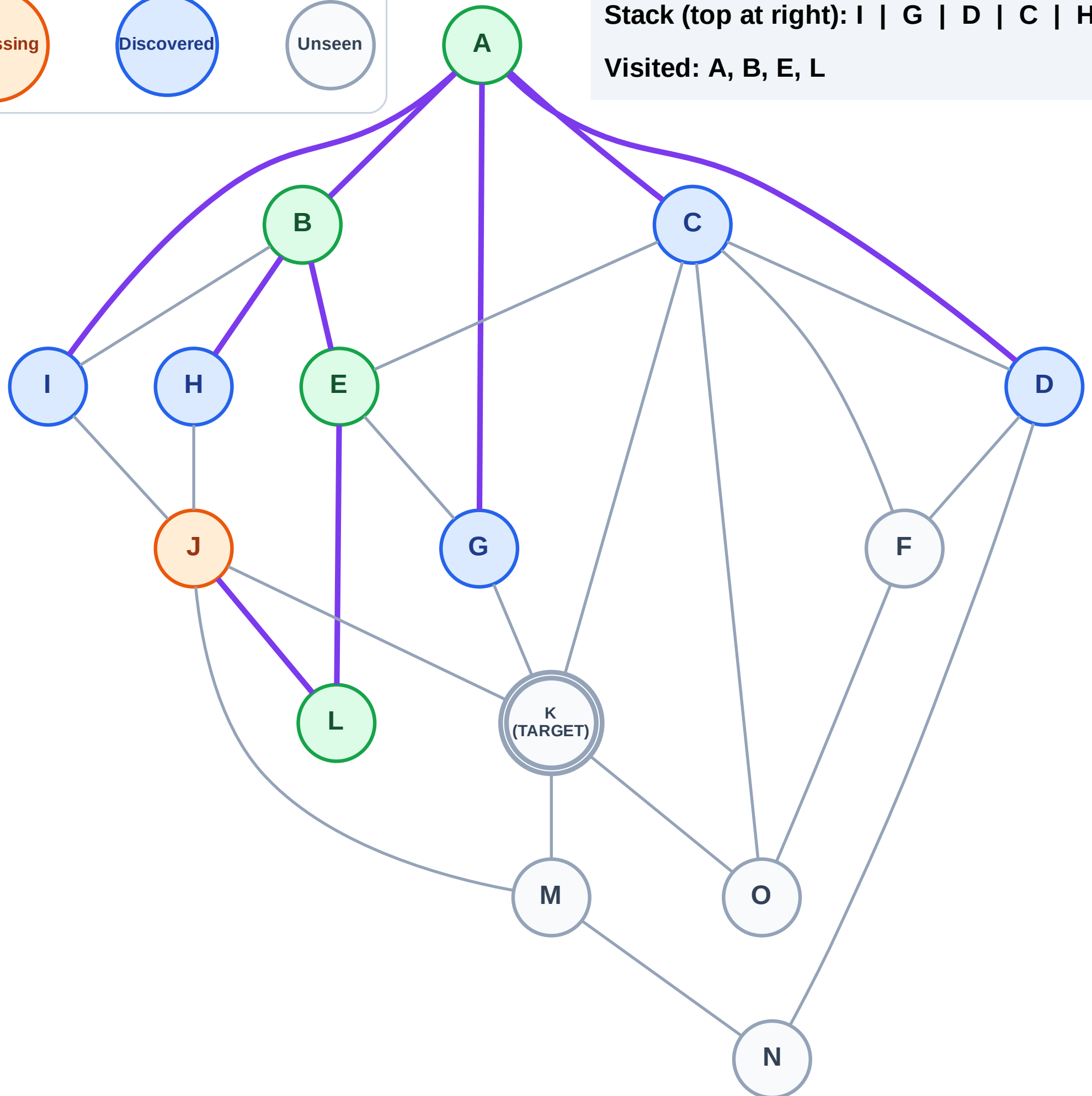
Processing

Discovered

Unseen

Stack (top at right): I | G | D | C | H

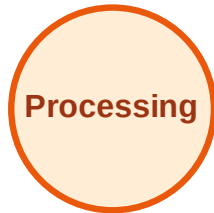
Visited: A, B, E, L



DFS Traversal

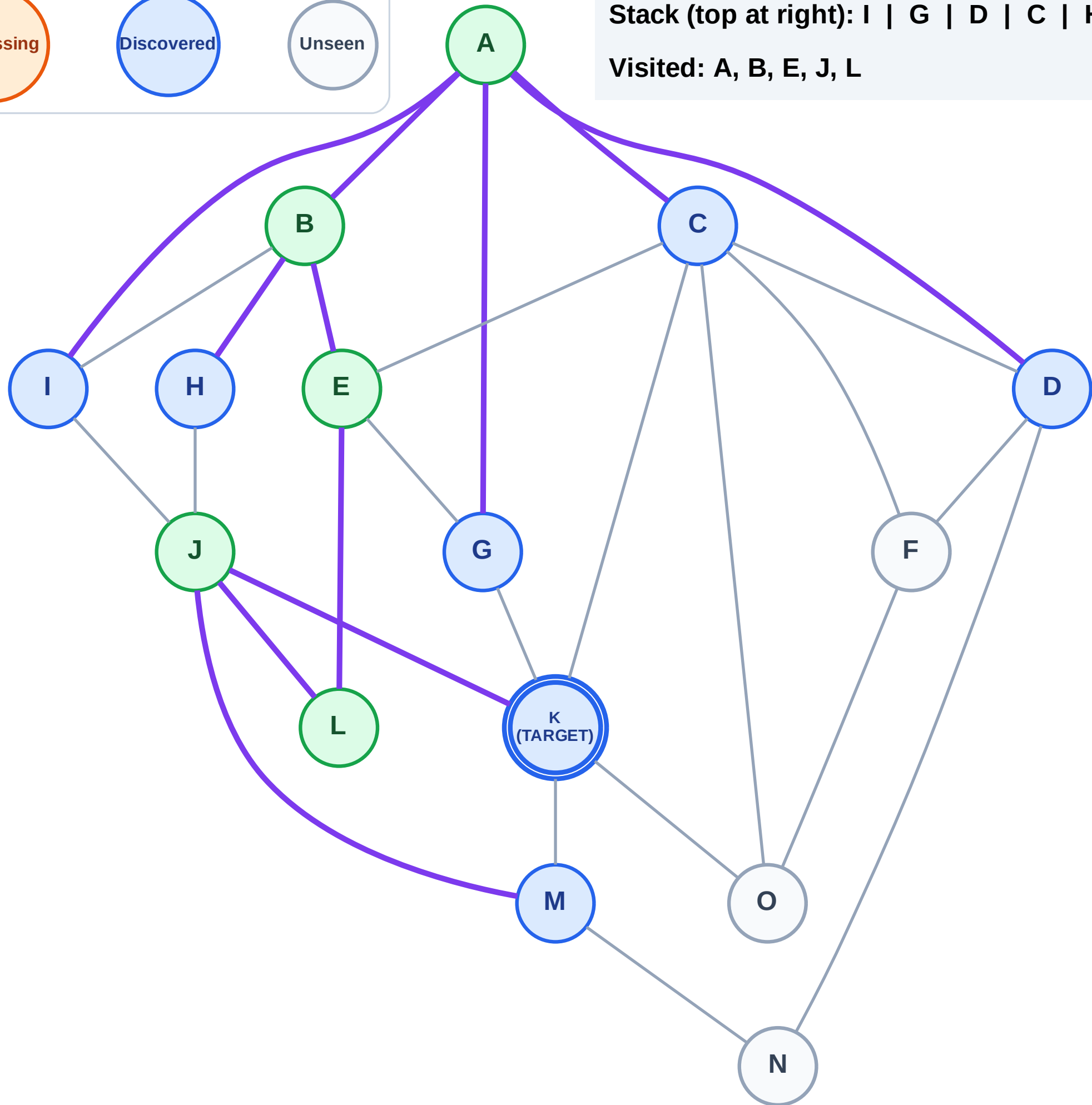
Step 11: Discover M, K from J

Legend



Stack (top at right): I | G | D | C | H | M | K

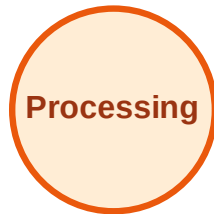
Visited: A, B, E, J, L



DFS Traversal

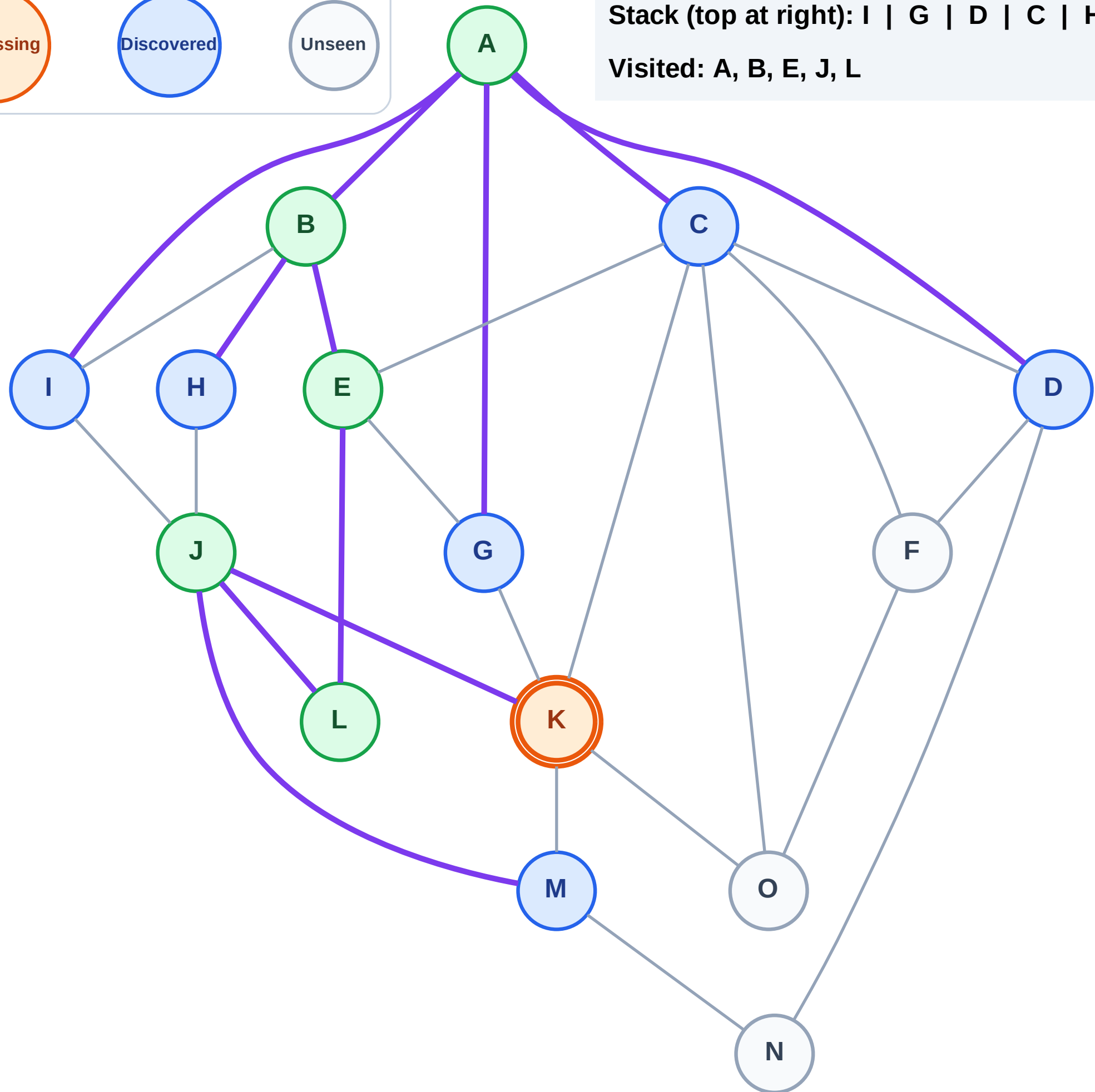
Step 12: Process node K

Legend



Stack (top at right): I | G | D | C | H | M

Visited: A, B, E, J, L



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → E → L → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | D | C | H | M
Visited: A, B, E, J, K, L

